Higher Order Sliding Mode Control and PREL Sliding Mode Control strategies For Urban Platooning

DISSERTATION REPORT

Submitted in partial fulfilment of the

requirements for the award of the degree

of

MASTER OF TECHNOLOGY

in

ELECTRICAL ENGINEERING

(With specialization in Systems and Control)

by

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JUNE 2021

I. CANDIDATE'S DECLARATION

I hearby declare that the dissertation work which is presented here, entitled ***Higher Order Sliding Mode Control and PREL Sliding Mode Control strategies For Urban Platooning*** , submitted in partial fulfilment of the requirements for the award of the Degree of **Master of Technology** in Electrical Engineering with specialization in System & Control at Department of Electrical Engineering, Indian Institute of Technology, Roorkee. I also declare that I have been doing my work from August 2020 under the supervision and guidance of **Dr.Sohom Chakrabarthy and Dr Arnab Dey, Assistant Professor, Department of Electrical Engineering, Indian Institute of Technology Roorkee**. The matter presented in this dissertation report has not been submitted by me for award of any other degree of institute or any other institutes.

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II. CERTIFICATE

This is to certify that the above statement made by the candidate is true to best of my knowledge and belief.

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ABSTRACT

The advancement in automation have propelled the technology of multi-agent systems which are used for various task like disaster management, obtaining the source of radiation leak, platooning, various military purposes. Automation in cars have also emerged which has lead to achieve an exceptional technology which is the Self-Driving car. The merging of Multi-agent system with the Self-Driving car results in autonomous platooning in which each agent will be communicating with each other to perform the task.

Platooning is one of the efficient methods for preventing traffic congestion and performance efficiency. The coming future would be propelled by electric vehicles and self-driving cars which drags the attention of researchers to investigate the automation in platooning. In this particular dissertation we would be discussing about the Higher order Super twisting Sliding mode control (SMC) and Sliding mode control with power rate exponential law for obtaining a robust, accurate platooning which can be considered for urban areas.

In the first chapter of the dissertation we would be discussing about multi-agent systems,self driving car and platooning. The second chapter would be mentioning about how various control approach and other algorithm were used in achieving platooning and also the reason for choosing higher order SMC and SMC using PREL over other controllers. Then mathematical modelling of the vehicle dynamics, theory and Application of PID, conventional SMC, Super twisting SMC and PREL would be discussed. A comparative study of all the controller would be mentioned in the sixth chapter in which the robustness of all the controller would be tested. The results of all the controllers are obtained through Matlab Simulink model.

ACKNOWLEDGEMENT

It is my desire to offer my sincere thanks to all those who have contributed in completion of the thesis work.

I would like to take the opportunity to express my sincere gratitude towards my supervisors Dr Sohom Chakraborthy Assistant and Dr Arnab Dey Assistant professor Department of Electrical Engineering Indian Institute of Technology Roorkee for their constant encouragement and motivation through out the duration of the thesis work.

I also express my sincere gratitude to Department of Electrical Engineering for providing me sufficient facilities throughout the year for the completion of the research work. I would also like to acknowledge research fellow Jagganath Samantharay who continuously supported and guided me throughout the completion of the project.

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Introduction Chapter-1

I. Multi-agent System(MAS)

A System containing a group of agents sensing the environment in which they are placed and interacting the information with each other for accurate decision making in order to complete a particular task. Multi-agent system emerged as they could accurately perform a particular task which is very much difficult or nearly impossible for one agent to perform for example finding a source of chemical leakage in a plant through odour intensity which is very difficult for one agent to locate the source as it has to search every area of the plant. In Multi-agent system there can be leader agent which leads the agent or all the agents perform the task without any leader. To perform a particular task instead of using a single agent which can consume lot of time or can consume more energy more than one agent of less bulky in nature can be used to improve efficiency and time consumption.

The agents in this particular system communicate its states or other experience obtained through sensing the environment to other agents through which a corporative control approach is achieved. The control approach is used for decision making allowing the agents to co-ordinately perform a task for achieving the goal of the particular system. The MAS is classified in two types

1. Homogeneous structure:-
   In this the agents have the same internal architecture referring to sensor functionalities, mechanics, parameters and goals.
2. Heterogeneous structure:-
   In this the agents can have different structures, task, actions based on the requirement for a particular environment. For example a leader agent may be a mobile robot where as other agents might be a swarm of small UAVs.

MAS is used in verity of application due their following benefits

1. Due to parallel communication and unsynchronized process an increase in efficiency and speed in the performance is achieved.
2. High level of redundancy and robustness are achieved as even failing of one or more agents does not affect the system.
3. Due to the use of many small individual agents the system is cost effective than a single centralized bulky agent.

4. As the agents can be added or removed flexibility, scalability is one of main advantages of MAS.

MAS can be widely applied in many areas like tracking, space exploration, search and rescue, traffic management, Supply-chain, surveillance. Due to these wide areas of applications multi-agent systems has emerged as a solution for modern society complications.

II. Autonomous Vehicle And Platooning

Self-Driving car is one of the most advanced automation system that is been developed by researchers extensively using advanced control system, Artificial Intelligence. A self driving car consist of many sensors like Lidar, IMU, Position encoders, for distance covered and environment scanning, ultrasound sensors for distance measurement between the car and obstacles. Most of the present self-driving cars are changing to camera or stereo vision which performs all the tasks mentioned above. The safety measures of self-driving car is one of key factors which is been looked upon and more improvement in this field is performed by applying more efficient sensors and controllers. As artificial intelligence is striving it is been applied with efficient controllers to obtain level 4 to 5 self-driving car meaning nearly autonomous and full autonomous in nature.

The combination of multi-agent system and self-driving cars is known as platooning. Platooning is a task in which each vehicle should follow its predecessors with a maintained distance. Platooning is performed to reduce traffic congestion, accidents and for less energy consumption. In platooning a leader agent or car is independent to the rest of the platoon and it is the only agent which decides the behaviour of the platoon. In platooning each agent communicates their position, velocity, parameters and environment behaviour to have a string stable platoon. The communication between each agent, efficient control and sensors are the deciding factors of a robust platooning.

Literature Survey Chapter-2

1. Introduction

This chapter contains the literature survey on Sliding mode Control and its applications in platooning. Section 2.2 contains the various recent sliding mode control strategies evolved which has been used for linear and non-linear system. Section 2.3 contains various Control strategies used for achieving various goals of platooning problem like string stability, formation and maintenance of the platoon of robots, vehicles. Section 2.4 contains various sliding mode control strategies and adaptive sliding mode control strategies used in platooning problem.

Section 2.2 Sliding mode Control Strategies.

The Sliding Mode Control scheme is a robust control method which was first used by G.N. Nikolski for Ship stabilization problem. Since then many innovative sliding mode techniques and its application evolved with rising constraints on each technique of SMC. The basic SMC which where formulated are first order SMC, Integral SMC, Terminal SMC. Each of these had their own limitations so various methods of using SMC evolved which looks into constraints like chattering effects, miss-Matched uncertainties, Finite time convergence, Non-linearity etc. Here we are going to discuss about only the recent Sliding Mode Control techniques which solves the constraints. In [1] a robust sliding surface is proposed for controlling the speed of the electromechanical plant. In this the SMC is generated with the PID sliding surface It is used for robustly track the trajectory with considering the uncertainty, Non-linear dynamics and external disturbances. The chattering problem is also overcome due the continuous hyperbolic function of the sliding surface. But In this research the finite time convergence of the state is not analysed and the analysis that the proposed sliding surface reaches zero not only for the linear system but also for non-linear system is not analysed so to achieve a finite time convergence of state terminal sliding mode control (TSMC) can be used but there is a singularity problem which can be removed through non-singular TSMC which is discussed in [2]. In [2] the research is based on achieving finite time tracking control (FTTC) for the rotor position in the axial direction of a nonlinear thrust active magnetic bearing system. In this study a Robust non-singular TSMC with hermit neural network is introduced which helps in predicting the uncertainties and accordingly a control signal is generated. The performance of RNTSMC is analysed with TSMC and NTSMC. Here using RNTSMC the bound of uncertainties need not be known in advance. As terminal Sliding mode control is an algorithm

which make states reach their goals in finite time it was used along with integral sliding mode control algorithm which gives a path of continuous tracking control and also there is a minimum chattering effect. The reach research for gaining non-singularity, fast convergence rate, small steady-state errors a ITSMC integral terminal sliding mode control was proposed which is combined with the adaptive law which is compensates the system uncertainties and actuator faults [3]. The above was used for linear system so to obtain a control algorithm similar to ITSMC the researchers proposed [4] an Integral Terminal Sliding mode control for non-linear systems which is combined with the back-stepping technique. The ITSMC checks the mismatched uncertainties and also the convergence rate is also analysed which is faster than the NTSMC. But the above ITSMC did not guaranty that the sliding surface will reach to zero and stays in it for infinite time. So to obtain the guaranty the research based on a strategy which involves the producing third order integral sliding mode control dedicated dedicated to motion tracking control of a piezoelectric-driven nano-positioning system. The proposed controller completely eliminates the chattering effects, achieves a finite-time convergence, and produces a higher sliding mode precision. The model uncertainty involving the hysteresis effect is estimated by the perturbation estimation technique [5].

Section 2.3 Control System Strategies on platooning

The Autonomous platooning attracted many researchers to apply different control algorithms along with neural-networks to obtain a robust platoon of vehicles. Mostly researchers applied linear platoon dynamics but as platooning was not only based on velocity, position dynamics of the agents it is dependent on the non-linear dynamics and kinematics of the agents also so many researchers started applying non-linear dynamics in the control problem to remove most of the constraints present in the platoon. The researches which can be analysed are [6] In this research the safety and fuel efficiency is also considered as a constraint and a two control layer structure is proposed with the information of road topography. A dynamic programming is computed to obtain fuel-optimal speed profile for the entire platoon. A distributed Model Predictive Control of vehicle is presented for real-time control of vehicle. Another research which similar to [6] is [7] in which a Model Predictive Control is used to obtain an optimal velocity of the whole platoon and a triple-step adaptive control is proposed to obtain a corporative control in the multi-agent system. These proposed control consider the goal of string stability and optimal fuel consumption as goals to be achieved. Another MPC based optimal control is also be reached upon in which the main focus was the string stability and collision avoidance[8]. In [8] two step distributed MPC control structure is used to solve the

optimal control problem over two vehicle structure using the communication between the vehicles. The proposed DPMC minimizes the cost function that represents the constraints in non-iterative manner. The control structure is string stable as it involves collision free properties. The above research did not focus on sensor uncertainties which can be analysed and solved referring to [9] in which an optimal control problem of the vehicle platoon formation and control has been investigated. . Here limitations of sensors and actuator such as throttle and braking delay is involved. These limitations and other constraints are combined to form a cost function through which a control signal is obtained which should also contribute for string stability in the platoon. In platooning communication system is also a very integral part which should not be damaged as it completes the network of the vehicle. [10] Is based on reducing the stress on the communication system due to excessive parameter to be communicated between the agents using event triggered control in which an event-triggered is modelled and if violated the control signal is communicated with each agent. The string stability is also considered and CTH policy is used to control each agent. Another research based on many of constraints present in the platoon is proposed in [11]. In this a robust control in which the platoon model is considered with parameter uncertainty, equivalent communication delay, sensor delay. Here Lyapunov–Krasovskii functional Hinf controller is designed to obtain a state feedback control which ensures string stability and agent's stability. Here L2 string stability is proposed for the controller. The above researches did involve constraints but it was solved in one dynamic model of the platoon. This control approach can have more mathematical complexity so to separate the constraints [12] can referred in which MMS Multiple model switching structure is considered in which many of the uncertainties are separated as models and switching method selects the most appropriate robust controller in the loop considering the errors between vehicle dynamics and the multiple models of the vehicle. The results are compared with sliding mode control and Hinf controller.

Section 2.4 Sliding Mode Control System Strategies on platooning

Many Sliding Mode Control Techniques are applied individually or using other controllers as to minimize the uncertainties and to robustly control almost all the dynamics of the vehicle and the platoon model. The dynamics of the first order sliding surface is generally taken as a discontinuous function which causes chattering so to counter this an improved first order sliding mode control theory was proposed in which there instead of using conventional sliding mode control dynamics which contains discontinuities a continuous dynamics of sliding variable is chosen[13]. The above research was based on linear dynamics of the platoon. But

even the non-linearity vehicle dynamics should also be considered which is proposed in upcoming researches.[14] a distributed sliding mode control is applied for non-linear dynamics system in which a topological sliding surface and reaching law is designed. The above two researches did not focus on disturbances and other uncertainties so instead of applying a direct sliding mode control an adaptive parameter is used which is constant time head away in which two sliding surface is defined for formation and Maintaining control[15]. The sliding surface based on the present agent is sometimes not sufficient to control the whole platoon as it does to only depend on present agent but also the previous agent. So to obtain a sliding surface based on present and previous agent [16] can be referred in which an adaptive coupled sliding mode control is used in which two sliding surfaces of previous and the following vehicle is used. The above researches where all based on first order sliding surface which when used for tracking problem can create a problem so a coupled integral sliding mode control is implemented using disturbance observance to make the platoon to a desired position with desired distance spacing [17-18].Another Adaptive controller is been discussed in [19] which uses Distributed Adaptive Integral sliding mode control in which the disturbances are estimated using the lower and upper bounds. An algorithm is also presented here to change the sliding surface mechanism which reduces the chattering disturbance. But uncertainties which are separated can be combined as one uncertainty which is been discussed in [20] which is based on based on Distributed Adaptive Sliding mode Control using integral sliding mode Control approach but here the uncertainties are been present in the acceleration which compensates other disturbances. Two adaptive control law are generated with two policies one is TCTH and the other is MCTH. Till now the research was based on the know non-linearity but there is a possibility that there can non-linearities in platoon dynamics which is not known so to predict the non-linearities [21] can be referred in which CNN based distributed Adaptive Control law is taken in which the CNN is used to obtain the approximate non-linearity of the vehicle system and an adaptive method with ISMC is used to obtain a controller parameter with TCTH and MCTH methods are used. Till now the discussion was on making the agents reach their goal by directly controlling the dynamics of the vehicle but there is possibility in which the mission goal is converted to behaviour of the vehicle or dynamics of the vehicle which is discussed in [22] in which an adaptive fuzzy control and terminal sliding mode control is applied to make a platoon reach a goal avoiding obstacles. The mission control is changed into a behavioural control of the vehicle and the goal is achieved by the vehicle. Till now we have not gone into faults on different components of vehicle which can affect the platoon control like actuator faults. [23] Discuss on how to obtain robust control even having these faults. In this research the fault

tolerant based control is used in which actuator faults using input quantization is generated and using radial basis function neural networks the non-linearties are approximated and using an adaptive PID based Sliding mode control a control input is generated. The string stability and non-zero-initial spacing error is also used. But even though faults are used time delay or lags are not considered in [23] so for obtaining the controller compensating the time delay or lags referring [24] is sufficient. In [24] a particle approach of first order sliding mode controller using CTH policy is applied in which time delays in the model and compensation of lags in string stability is being done. The analysis on both time delay and lags is also done. The above researches were based on a reaching law which is based on a discontinuity and sometimes directly proportional to the sliding surface. A new reaching law is discussed in [25] in which a two level controller is used to control all the possible dynamics of the platoon system. A PRERL power rate exponential reaching law based sliding mode control is used to obtain torque input and a low level based PID controller is used to compensate the delay in the actuation system. Both the control output is sent to the wheel dynamics and platoon dynamics. Another Fuzzy control method is been upon in which a hierarchical fuzzy logic with sliding mode control is used in which the first layer is used for spacing control with acceleration and velocity control. The sliding mode controller is used for adaptively compensate for disturbances and uncertainties in it. In the above researches the sliding variable is directly generated through the position and velocity errors. But a new sliding surface is generated using unconstrained transformed error in [27] which is then used for decoupled sliding mode control for obtaining Prescribed Performance control ensure strong string stability.

Conclusion

In this chapter the literature Survey is done on recent Sliding Mode Control, Optimal Control and PID control applications on platooning. In the above literature survey it can be seen that in SMC control platooning the faster and finite time convergence of the state are not considered though most of the uncertainties are compensated and the chattering effect is also well governed. The method of power rate reaching law and higher order super-twisitng algorithm are not considered which make sure that the states reach their goal in finite time and at much faster rate. So the platooning based on Higher order Sliding mode control and power rate exponential reaching law is to looked upon.

The model of the vehicle is obtained through the dynamics of the bicycle model shown in figure 1. The dynamics of the vehicle are used to obtain the longitudinal and lateral control of the vehicle. In the modelling all the vehicle which is also the agents contain same model and the number of vehicles considered are N. the dynamics of the jth vehicle is given as

$$\dot{U}_j = \frac{V_j r_j}{M} - Fg + \left(\frac{U_j^2(FK_l - K_d)}{M}\right) + \frac{c_{ft}(V_j + ar_j)}{U_j M} d_j + \frac{f_j}{M} \quad (1)$$

$$\dot{V}_j = \frac{1}{M}\left(-MU_j r_j - \frac{c_{ft}V_j}{U_j} - \frac{c_{rt}V_j}{U_j} + \frac{bc_{rt}r_j}{U_j} - \frac{ac_{ft}r_j}{U_j} + c_{ft}d_j + f_j d_j\right) \quad (2)$$

$$\dot{r}_j = \frac{1}{j_z}\left(\left(\frac{bc_{rt}V_j}{U_j} - \frac{ac_{ft}V_j}{U_j}\right) - \frac{b^2 c_{rt}r_j}{U_j} - \frac{a^2 c_{ft}r_j}{U_j} + ac_{ft}d_j + af_j d_j\right) \quad (3)$$

$$\dot{X}_j = U_j \cos(\Psi_j) - V_i \sin(\Psi_j) \quad (4)$$

$$\dot{Y}_j = U_j \sin(\Psi_j) + V_i \cos(\Psi_j) \quad (5)$$

$$\dot{\Psi}_j = r_j \quad (6)$$

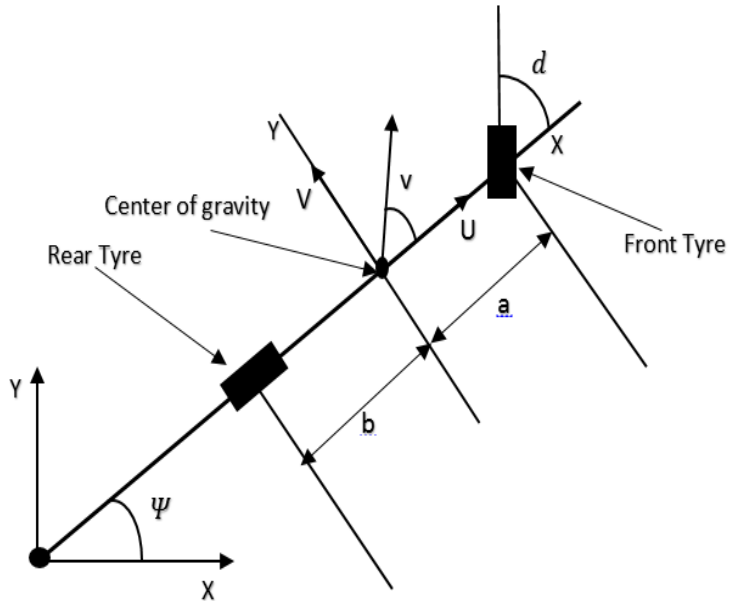


Fig. 1. Bicycle model of a vehicle

where U_j, V_j, and r_j are longitudinal, lateral and yaw rate, respectively. Ψ_j is the yaw angle. f_j is the traction force between tyre and the ground and d_j is the steering angle. (X_j, Y_j) is the coordinate of the center of gravity of the vehicle. The remaining constant parameters are

presented in Table I.

TABLE I: PARAMETER OF THE MODEL

Symbol	Description	Value
M	Mass of the vehicle	$1750Kg$
F	Rotating friction coefficient	0.05
j_z	Inertial Moment along z-axis	$2350Kgm^2$
a	Distance between front tyre and center of mass	$1.05m$
b	Distance between rear tyre and center of mass	$1.63m$
c_{ft}	Cornering stiffness of front tyre	$\frac{150000N}{rad}$
c_{rt}	Cornering stiffness of rear tyre	$\frac{90000N}{rad}$
K_l	Lift obtained due to aero dynamics	$\frac{0.005Ns^2}{m^2}$
K_d	Drag obtained due to aerodynamics	$0.25\frac{Ns^2}{m^2}$

The model above is a non-linear model and to control the dynamics the model is linearized through Taylor series method. A linearized model of equation (1-3) is obtained keeping the equilibrium points of the jth as U_{0j} is the initial velocity of each vehicle and $V_{0j} = 0, r_{0j} = 0$. The linearized longitudinal model is presented found as

$$\dot{U}_j = \frac{2U_{0j}(FK_l - K_d)U_j}{M} + \frac{f_j}{M} \qquad (7)$$

Similarly, the lateral velocity dynamics is found as

$$\begin{bmatrix} \dot{V}_j \\ \dot{r}_j \end{bmatrix} = \begin{bmatrix} -\frac{c_{rt}+c_{ft}}{MU_{0j}} & \frac{bc_{rt}-ac_{ft}}{MU_{0j}} - U_{0j} \\ \frac{bc_{rt}-ac_{ft}}{j_z U_{0j}} & \frac{-(b^2c_{rt}+a^2c_{ft})}{j_z U_{0j}} \end{bmatrix} \begin{bmatrix} V_j \\ r_j \end{bmatrix} + \begin{bmatrix} \frac{c_{ft}}{M} \\ \frac{ac_{ft}}{j_z} \end{bmatrix} d_j \qquad (8)$$

The linearized $\dot{X}_j$ and $\dot{Y}_j$ are represented as

$$\dot{X}_j = U_j - V_j\Psi_j \qquad (9)$$

$$\dot{Y}_j = U_j\Psi_j + V_j \qquad (10)$$

The problem here is to design the control inputs i.e. tracking force, f and the steering angle, d such that U_j tracks the reference longitudinal velocity U_{ref} and the lateral reference trajectory Y_{ref}, respectively. Using the above mathematical model controller such as PID, conventional sliding mode control, sliding mode control using PREL and Higher order with relative degree 1 and 2 super-twisting sliding mode control are proposed for the formation of the platoon and to follow the reference path trajectory. The robustness of each controller would be discussed in the following chapters and a comparative study would also be presented. The results of the controllers designed on the above model will also be discussed.

Design of PID controller Chapter-4

Proportional Integral Derivative Controller one of the most simple and efficient controller still used in process, robotics and various industries. It is still used as its complexity is less and satisfactory results can still be obtained but it can only be used in linear model. There is one disadvantage of PID which is that it does not robustly reduces disturbances. The PID controller must always be tuned properly till the results is accurate. PID control is closed loop control which uses feedback from the measurement and checks it with the reference which results in error. This error is passed through PID controller which then outputs a control input for the system or plant. PID controller works on three responses.

1). Proportional response

This type of response is achieved by incorporating a proportional gain term Kp with the error which also acts as the ratio of the response to the error. If an error term is of 5 and the gain selected as 15 then output of the controller would be 75 so if error increases the response of the controller increases which help in converging the error to zero at a faster rate. But if the gain Kp is chosen for a larger value than the required then oscillation starts appearing which increases if Kp is increased. If the oscillation increases then the system will be out of control due to instability. If only Kp term is used then the controller is called proportional controller.

2). Integral Response

The function of the integral is to add the present error term to the previous term. The integral action forces the steady state error to decrease till it converges to zero. The integral action is accompanied with a proportional gain Ki which should be chosen according to controller response. If the proportional gain is increased inappropriately then it will saturate the controller before the steady state error reaches zero.

3). Derivative response

It is the rate of change in the error. As the integral response forces the controller to keep the steady state error zero an overshoot in the process variable is attained which can cause a jerk in the system to avoid this phenomenon and to accurately settle the error to zero an derivative action is required which increases the control out if there is an increase in the change of the error. The derivative controller is also accompanied with a gain Kd which should be chosen according to the requirement. If Kd is increased unnecessarily then settling time increases and

as the derivative is sensitive to noise increasing its gain would result in unstability of the system.

The mathematical formula of the PID controller is

$$u = K_p e + K_i \int e + K_d \frac{de}{dt} \qquad (11)$$

Where e is the error and u is the control output.

Thus PID controller is accurate and safe controller which can drive the system for attaining the desired process variable without any damage to the system and to have accuracy in the controller response PID controller is used instead of PI or PD controller which have more limitations.

1. Application of PID controller for longitudinal platooning

a) Leader Vehicle

For the leader vehicle a reference velocity is given so that the platoon would have a desired velocity.Through the application of (11) The PID controller is accommodated with the longitudinal dynamics of the vehicle (7). The error is considered as the difference between U_{ref} and U_1 which is then passed through the PID controller. The controller outputs the f_j as the control output for the system. The equation is shown below.

$$e = U_{ref} - U_1 \qquad (12)$$

$$f_1 = K_p e + K_i \int e + K_d \frac{de}{dt} \qquad (13)$$

The above control output makes sure that the velocity of the leader is reached to the desired velocity. The controller is applied as expressed in figure 1 below. The output of the system that is U_1 is compared with the set point through mux and scope.

b) Follower Vehicle

For longitudinal control of follower vehicles the error e_j should be considered as difference between desired safe distance spacing between vehicle D and $X_{j-1} - X_j = x_d$ which represents the spacing between each vehicle. The controller output f_j where $j = 2..N$ for follower is derived as

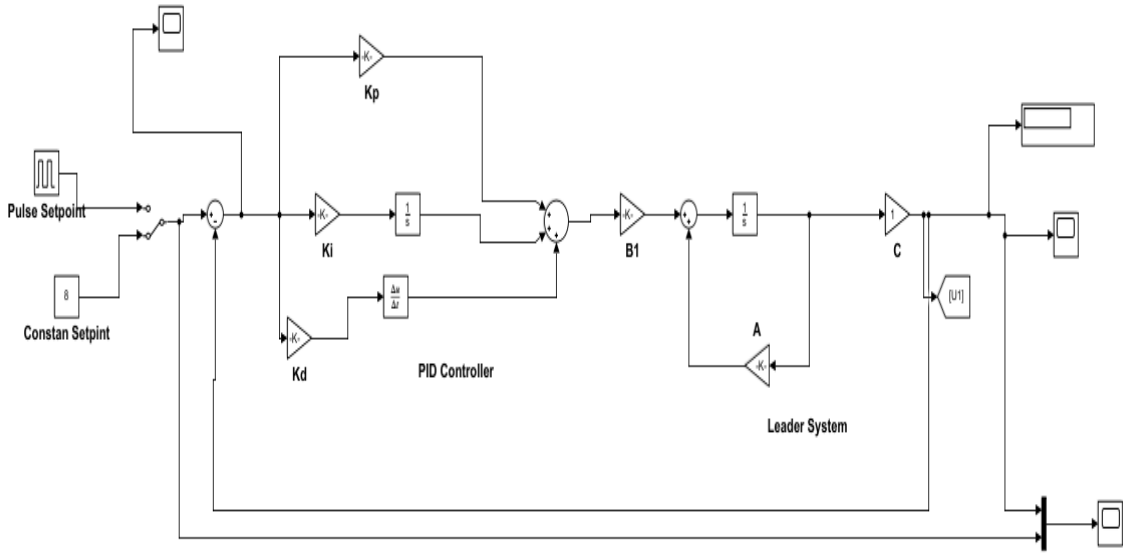


Figure 1 PID Leader Control

$$D = d + t_h U_j \qquad (14)$$

Where d is the desired distance between each vehicle or agent when they are at rest, t_h is the constant time head away chosen for string stability in the platoon. The error is written as

$$e_j = D - x_d \qquad (15)$$

$$e_j = d + t_h U_j - X_{j-1} - X_j \qquad (16)$$

$$f_j = K_p e_j + K_i \int e_j + K_d \frac{de_j}{dt} \qquad (17)$$

The control output ensures that the agents maintain the desired distance and also reach the velocity of the leader vehicle. The controller maintains the stability of the system and as the system contains time head-away constant the formation of the platoon is also means there will be no collision and sudden large distance between each agent which is also known as string stability. The following figure 2 express the application of the controller.

2. Application of PID for Lateral Control

The control of lateral velocity and yaw rate of the vehicle is known as the lateral control. For obtaining lateral control error $E_j = Y_{ref} - Y_j$ should be zero. Y_{ref} can be any path given to platoon to follow and here a sinusoidal reference path is considered. The following equations formulates the control output.

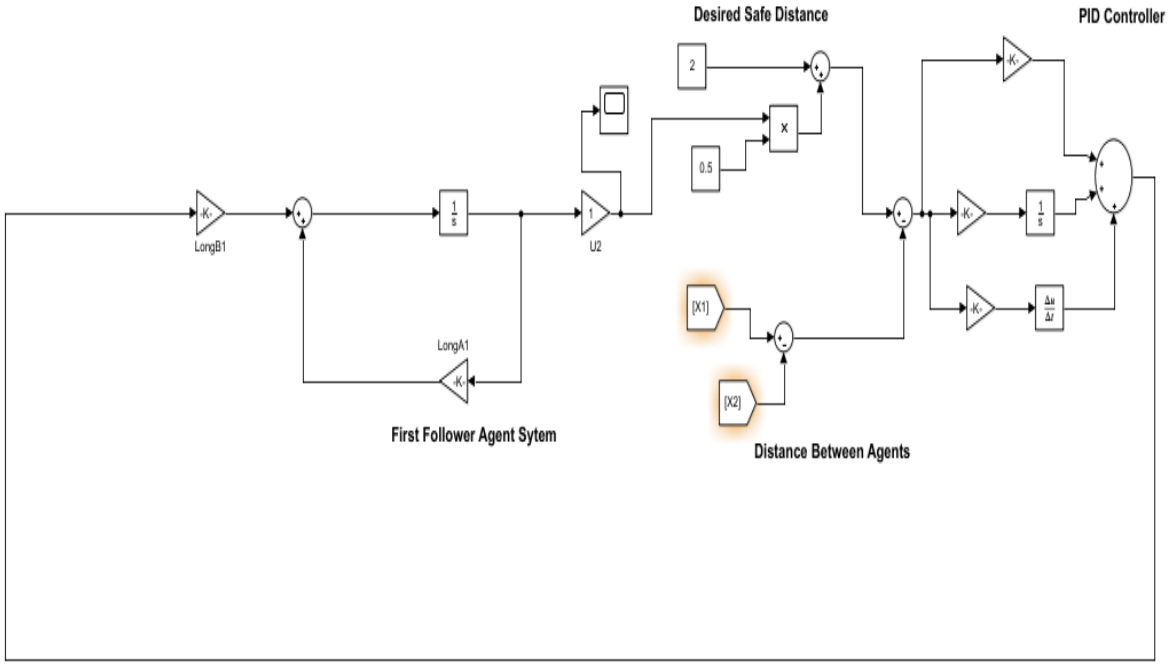


Figure 2 Follower controller PID

$$E_j = Y_{ref} - Y_j \qquad (18)$$

$$d_j = K_p E_j + K_i \int E_j + K_d \frac{dE_j}{dt} \qquad (19)$$

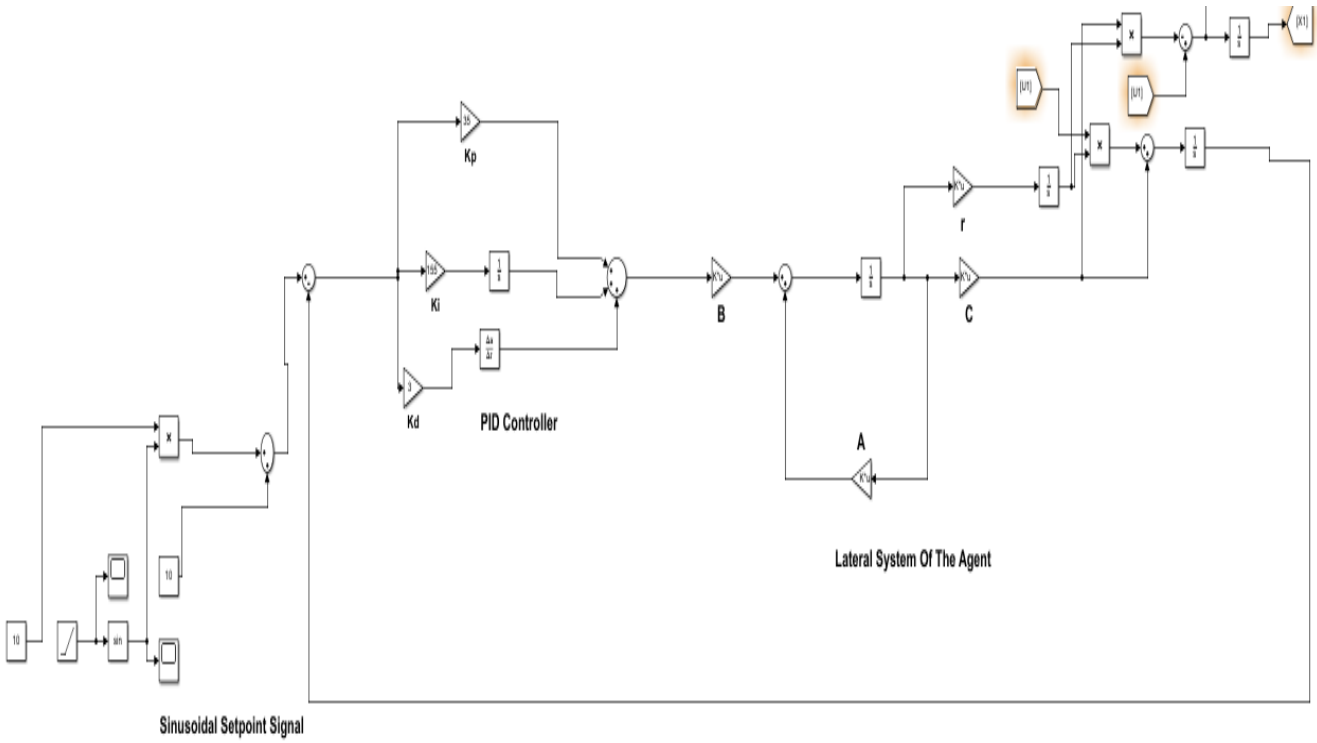


Figure 3 PID Lateral Control

The control output ensures the Y_j tracks Y_{ref} accurately so that the agents function like a platoon tracking a path. The lateral control is also called as steering control of the vehicle. Fig-3 shows the model of lateral control. The combination of both control a platoon formation along with path tracking is achieved. Even though the accuracy of the controller is good it is susceptible to disturbance and it is slower in nature. So a robust controller is needed for improvement in the platoon formation and path tracking. The results of the PID controller will be discussed in the coming chapter.

Design of Sliding Mode Controller Chapter 5

I. Conventional Sliding Mode Controller

Sliding Mode control is one of the non-linear robust control which outputs a discontinuous control signal which forces the system to reach a desired process value. In Sliding mode control a sliding variable is defined according to the requirement and a control signal is derived through the dynamics of the sliding variable which makes the sliding variable zero which is known as the sliding surface in finite time. As the Sliding surface is reached the system staes slide along the surface till the states reach their desired values asymptotically or in finite time.

The sliding mode control is of two parts which is designing a sliding variable which satisfy the process needs and specifications, designing the control signal through the dynamics of the sliding variable which is equated with a discontinuous signal. The discontinuous signal effects the dynamics of the variable which in turn effects the sliding variable. The control signal designed should maintain the desired sliding mode dynamics which is the reachability condition. Generally the sliding variable is defined as s and its dynamics $\dot{s}$ which should satisfy the reachability condition mentioned below.

$$s\dot{s} \leq -\eta|s| \qquad (20)$$

Where η is constant positive parameter. The equation (20) must be satisfied in the sliding mode control to attain the stability and robustness.

1. Application of Sliding Mode Control for longitudinal Platooning

a) Leader Vehicle

The Leader vehicle is independent to the agents or follower and it is the only agent which decides the behaviour of the platoon. The leader vehicle is given a reference longitudinal velocity U_{ref} which is to be reached. The U_{ref} is the key component which effects the platoon nature. The sliding variable s is chosen as the error between reference and the leader velocity. Through the dynamics of the sliding variable a control signal f_1 is attained. The equation shows the derivation of the control signal.

$$s = U_1 - U_{ref} \qquad (18)$$

$$\dot{s} = \dot{U}_1 - \dot{U}_{ref} \qquad (19)$$

substituting (7) in (19)

$$\dot{s} = \frac{2U_{01}(FK_l - K_d)U_1}{M} + \frac{f_1}{M} - \dot{U}_{ref} \quad (20)$$

the dynamics of the sliding variable is defined as

$$\dot{s} = -Ksign(s) \quad (21)$$

where K is a positive proportional parameter determined for stable rate of convergence to zero. Substituting (21) in (20)

$$-Ksign(s) = \frac{2U_{01}(FK_l - K_d)U_1}{M} + \frac{f_1}{M} - \dot{U}_{ref} \quad (22)$$

the control signal is calculated as

$$f_1 = M\left(-Ksign(s) + \dot{U}_{ref} - \frac{2U_{01}(FK_l - K_d)U_1}{M}\right) \quad (23)$$

The control signal f_1 satisfies the reachability condition $s\dot{s} \leq -\eta|s|$ and $s = 0$ is achieved in finite time which proves that the error reaches zero in finite time. The f_1 ensures that the state U_1 reaches U_{ref} accurately, robustly with maintaining the stability of the dynamic system of the vehicle. The following figure reflects the model of the control algorithm obtained.

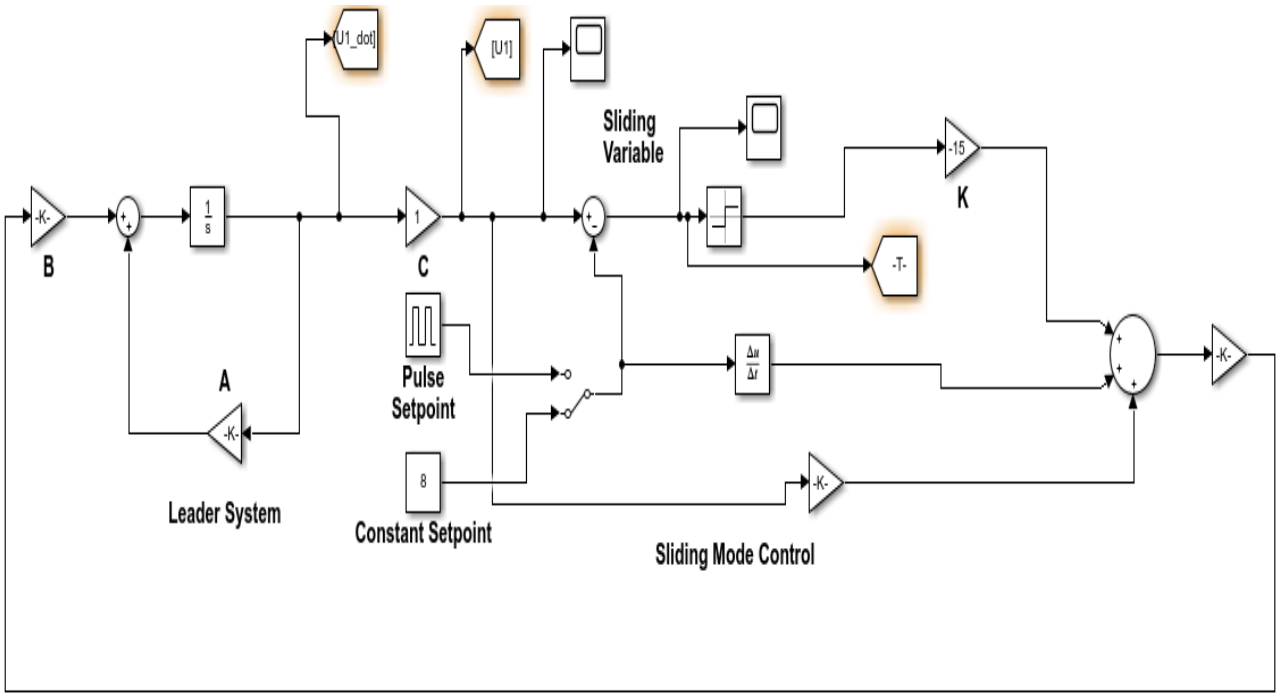


Figure 4 SMC Leader

The U_{ref} can be a velocity path like a pulse generator, sine wave or it can a constant set-point.

a) Follower Vehicle

For the follower vehicle the error e_j is considered as we considered in the PID. The sliding variable chosen is same as the error. Through the dynamics of the sliding surface and with a discontinuous function f_j where $j = 2..N$ for follower control is obtained. The desired safe distance is considers as $D = d + t_h U_j$. The equation is written as follows

$$S_j = d + t_h U_j - (X_{j-1} - X_j) \quad (24)$$

$$\dot{S}_j = t_h \dot{U}_j - (\dot{X}_{j-1} - \dot{X}_j) \quad (25)$$

substituting (7) in (25)

$$\dot{S}_j = t_h \left(\frac{2U_{0j}(FK_l - K_d)U_j}{M} + \frac{f_j}{M} \right) - \dot{X}_{j-1} + \dot{X}_j \quad (26)$$

The dynamics of the sliding variable is

$$\dot{S}_j = -Ksign(S_j) \quad (27)$$

Substituting (27) in (26) then

$$-Ksign(S_j) = t_h \left(\frac{2U_{0j}(FK_l - K_d)U_j}{M} + \frac{f_j}{M} \right) - \dot{X}_{j-1} + \dot{X}_j \quad (28)$$

$$f_j = \left(\frac{-Ksign(S_j) - \dot{X}_j + \dot{X}_{j-1}}{t_h} + \frac{2U_{0j}(FK_l - K_d)U_j}{M} \right) M \quad (29)$$

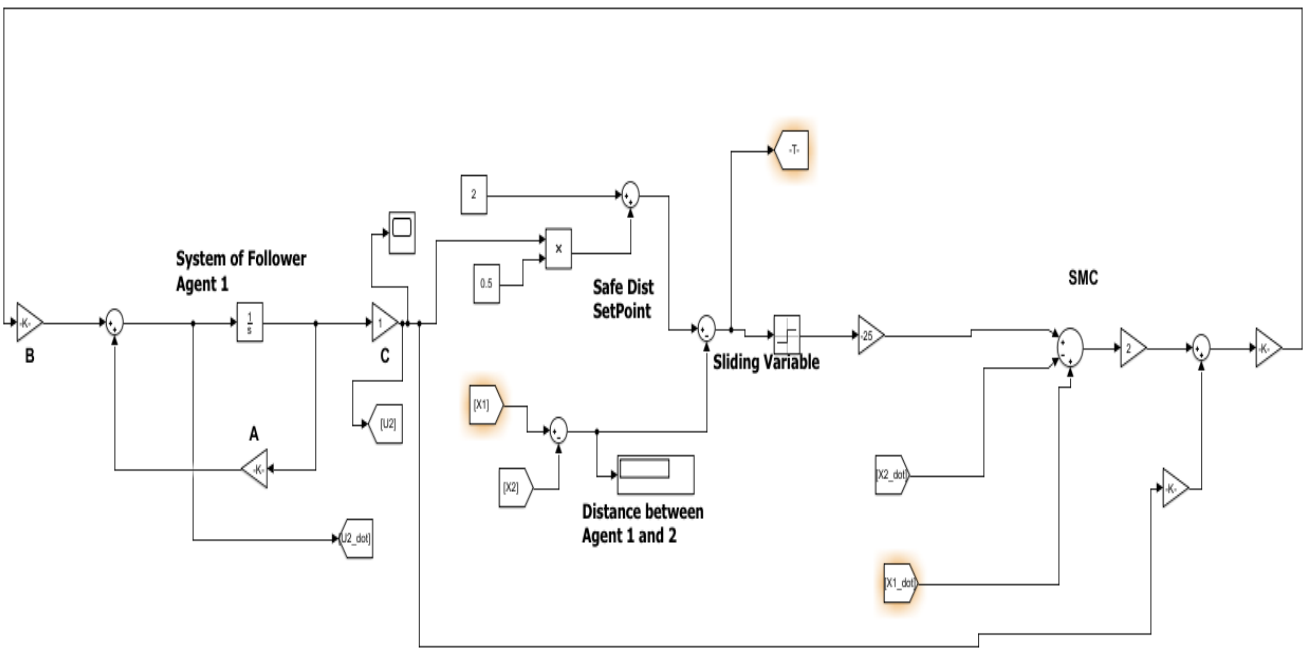


Figure 5 SMC Follower Agents

The above control signal forces the error to become zero in finite time as the condition $S_j\dot{S}_j = -\eta|S_j|$ is satisfied due to which sliding variable reaches zero in finite time. The control signal also forces the U_j from its present value to U_{ref}. Here the string stability is also maintained as constant time head away is used. The proof of the finite time reaching of error to zero for both leader and follower vehicles will be explained in the coming chapter. The model of the follower control is represented in the figure above

2. Application of Sliding Mode control for path Tracking

The Lateral control is performed by considering a sinusoidal reference path Y_{ref} through which the error between current Y_j where $j = 1..N$ coordinate is calculated which is represented as E the sliding variable is chosen as γ and through its dynamics the steering control or the lateral control is achieved. The sliding variable and the equation of deriving the control signal are as follows.

$$E_j = Y_j - Y_{ref} \qquad (30)$$

$$\dot{E}_J = V_j + U_j\Psi_j - \dot{Y}_{ref} \qquad (31)$$

$$\gamma_j = \dot{E}_J + lE_j \qquad (32)$$

$$\dot{\gamma}_J = \dot{V}_j + \dot{U}_J\Psi_j + U_j\,\dot{\Psi}_J - \ddot{Y}_{ref} \qquad (33)$$

the dynamics of the sliding variable is calculated as

$$\dot{\gamma}_J = -\Lambda sign(\gamma_j) \qquad (34)$$

where the Λ is a positive constant.l is also a positive constant which determine the convergence rate of error to zero. From equation (8) $\dot{V}$ is substituted in (33) and (34) is substituted in (33). The control signal d_j is calculated which is shown as

$$d_j = \frac{1}{\frac{c_{ft}}{M}}\Big(-\Lambda sign(\gamma_j) - U_j\dot{\Psi}_j + \ddot{Y}_{ref} - lV_j - U_j\Psi_j + l\dot{Y}_{ref} + \frac{c_{rt}+c_{ft}}{MU_0}V_j + \Big(\frac{bc_{rt}-ac_{ft}}{MU_0} - U_0\Big)r - \dot{U}_J\Psi_j\Big) \qquad (35)$$

The above control signal robustly make the system track the sinusoidal reference path. It also ensure the condition $\gamma_j\dot{\gamma}_j = -\eta|\gamma_j|$ is satisfied when the sliding variable reaches zero in finite time. Here the error reaches zero asymptotically due to the chosen sliding variable. So Y_j reaches Y_{ref} asymptotically. Due to the steering control and longitudinal control the platoon is

maintained in a path without deviation and the string stability is maintained. The sliding mode control is chosen as it is robust in nature, not effected by the disturbance much. The below figure describe the model of the path tracking control for the first agent.

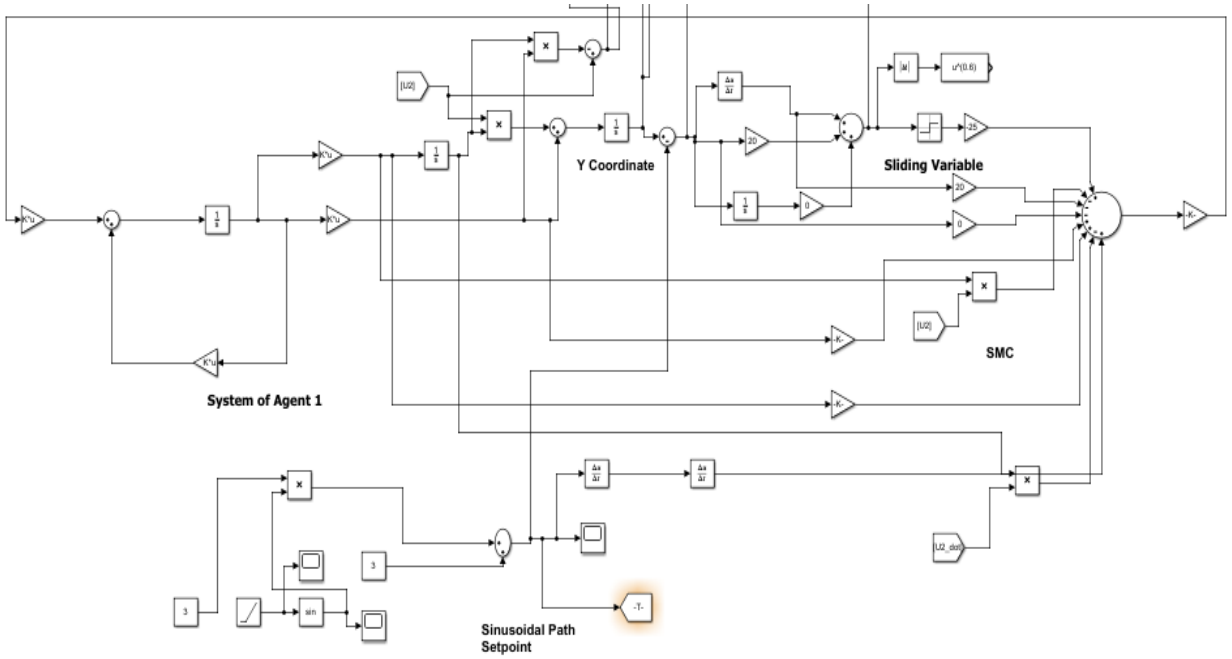


Figure 5 SMC Follower Agents

But there is problem in the conventional sliding mode control which is the chattering effect. The chattering effect is caused due to the discontinuous function used in the sliding variable and it can only be controlled by K and Λ here but even though a suitable value is included the chattering effect will not be negligible. The chattering effect is dangerous to the actuators of the system as it causes sharp oscillations in control signal and if suitable K and Λ is not chosen the actuators would get damaged immediately and cause collisions or an accident. So a more robust sliding mode control which eliminates the chattering effect is to be introduced.

II. Sliding Mode Control using Power rate reaching Law

1. Power rate reaching law

The power rate exponential reaching law is a one of the reaching strategy which is formulated as

$$\dot{s} = -K|s|^{\frac{p}{q}}sign(s)$$

It is chosen as the dynamics of the sliding variable chosen. Here K should be a positive parameter and $0 < \frac{p}{q} < 1$ condition should be applied. This reaching law ensures more faster

convergence of sliding variable to zero when it is very large and slower convergence when the sliding variable is near zero and this process is also applied in the convergence of the states to its desired location. Other than $K\ |s|^{\frac{p}{q}}$ also effects the dynamics of the sliding variable due to which the response of the sliding mode control is faster and the states reach their desired value in less time than conventional reaching law. Due to the characteristics of the law the chattering effect is eliminated. So by using this law in the SMC there would be not chattering effect which cause damage to actuators of the system. The reachability condition which is to be satisfied is $\dot{s}s \leq -\eta|s|^{\frac{p}{q}}|\ s|$.

2. Application of the power rate reaching law for longitudinal platoon

A Leader Vehicle

The leader vehicle of the platoon is the key agent as the follower agents are dependent on its velocity and accelerations. A reference speed profile U_{ref} is provided which is to be tracked by the leader agent. Let the longitudinal velocity of the leader be denoted as U_1. The sliding variable is chosen as

$$S_1 = U_1 - U_{ref} \qquad (36)$$

One gets

$$\dot{S}_1 = \dot{U}_1 - \dot{U}_{ref} = \frac{2U_{01}(FK_l - K_d)U_1}{M} + \frac{f_1}{M} - \dot{U}_{ref} \qquad (37)$$

The power rate exponential reaching law is represented as

$$\dot{s} = -\Lambda|s|^{\frac{p}{q}}sign(s). \qquad (38)$$

Where s is a sliding variable

From the equation (7) and (38)

$$-\Lambda|S_1|^{\frac{p}{q}}sign(S_1) = \frac{2U_{01}(FK_l - K_d)U_1}{M} + \frac{f_1}{M} - \dot{U}_{ref} \qquad (39)$$

Through the equation (12) the controller output is computed as

$$f_1 = M\left(-\Lambda|S_1|^{\frac{p}{q}}sign(S_1) + \dot{U}_{ref} - \frac{2U_{01}(FK_l - K_d)U_1}{M}\right) \qquad (40)$$

The above controller output (40) confirm that the reaching condition $\dot{S}_1 S_1 \leq -\eta|S_1|^{\frac{p}{q}}|\ S_1|$ is followed and the $S_1 = 0$ is reached in finite time and the error becomes zero in finite time too. The following figure represent the control model of the leader vehicle

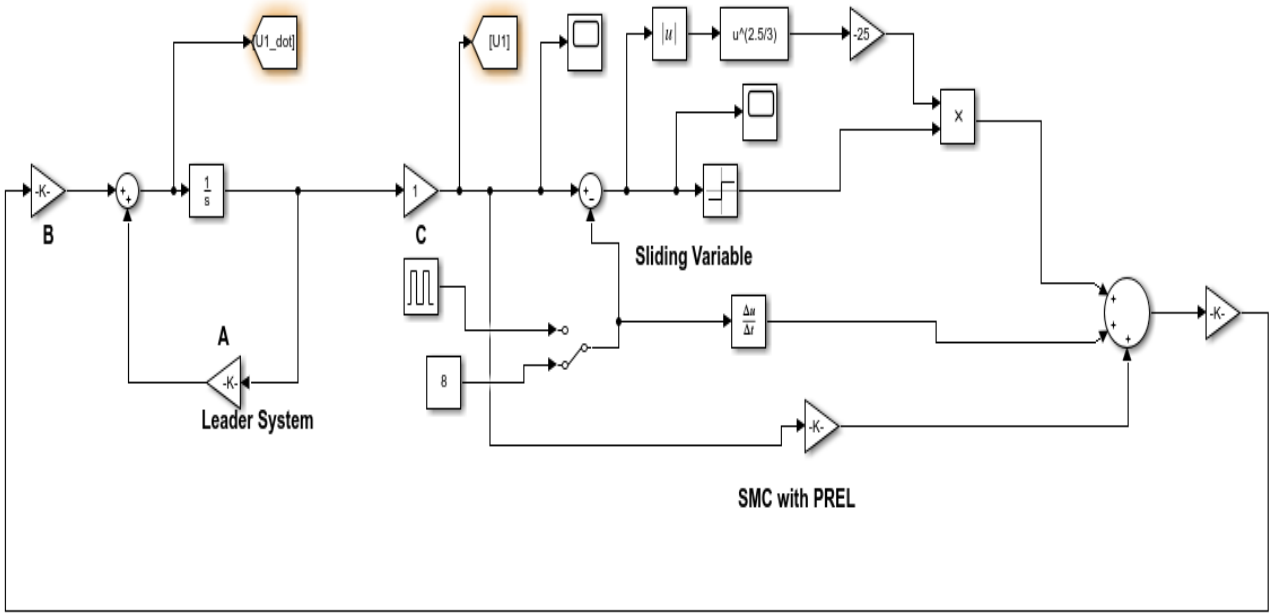


Figure 6 SMC PREL leader

B Follower Vehicle

For the formation of longitudinal platoon, we need to consider a desired safe distance between each agents which is represented as

$$D = d + t_h U_j \qquad (41)$$

where d is the desired distance between each agent when they are at rest and t_h is the constant time head away which is an adjustable parameter for increasing string stability in the platoon. Here $j = 2..N$. The distance between the agents is defined as

$$x_d = X_{j-1} - X_j \qquad (42)$$

The error between the (42) and (41) is called the spacing error which is represented as

$$e_j = D - x_d \qquad (43)$$

Substituting (42) and (41) in (43), it is found that

$$e_j = d + t_h U_j - X_{j-1} + X_j \qquad (44)$$

The sliding variable s_j is chosen as

$$s_j = d + t_h U_j - X_{j-1} + X_j \qquad (45)$$

$$\dot{s}_j = t_h \dot{U}_j - \dot{X}_{j-1} + \dot{X}_j \qquad (46)$$

equating(38) and (46), one finds

$$-\Lambda |s_j|^{\frac{p}{q}} sign(s_j) = t_h \dot{U}_j - \dot{X}_{j-1} + \dot{X}_j \qquad (47)$$

And through equation (7)

$$-\Lambda|s_j|^{\frac{p}{q}}sign(s_j) = t_h\left(\frac{2U_{0j}(FK_l-K_d)U_j}{M}+\frac{f_j}{M}\right)-\dot{X}_{j-1}+\dot{X}_j \quad (48)$$

Through equation (48) the control law is found as

$$f_j = \left(\frac{-\Lambda|s_j|^{\frac{p}{q}}sign(s_j)-\dot{X}_j+\dot{X}_{j-1}}{t_h}+\frac{2U_{0j}(FK_l-K_d)U_j}{M}\right)M \qquad (49)$$

The above control law ensure that the sliding variable satisfies the reaching condition $\dot{s}_j s_j = -\eta|s_j|^{\frac{p}{q}}|s_j|$ and allow the sliding variable converge to zero at faster rate in finite time. The model of this is represented in the figure below.

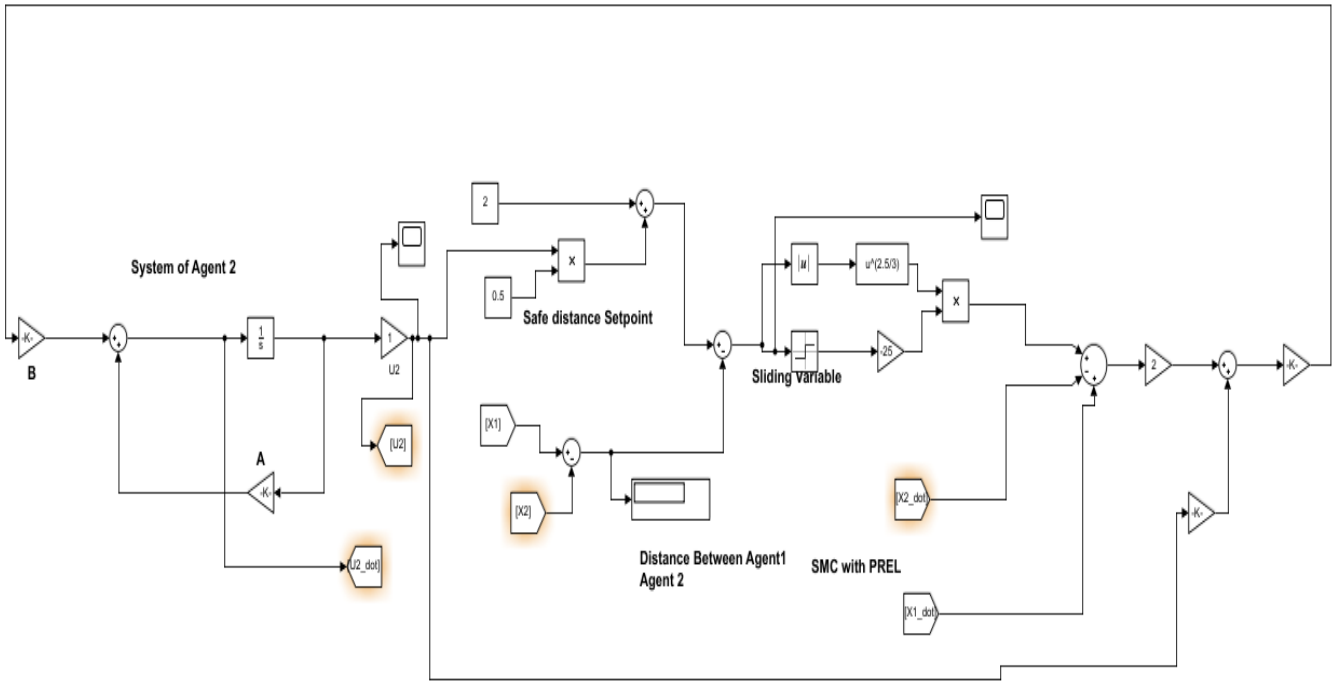


Figure 6 SMC PREL Follower Agent

3. Application of the power rate reaching law for Lateral Control

For lateral control a desired reference sinusoidal trajectory Y_{ref} is considered which all the agents should track without deviating from the platoon formation. The error is defined as the distance between the Y_{ref} and Y. For simplicity the $j = 2..N$ is omitted in the equation.

$$E = Y - Y_{ref} \qquad (50)$$

For following the reference trajectory, a control law should be implemented which converge the error to zero. The sliding variable is chosen with PID sliding surface where the both convergence time and steady state behaviour can be controlled. The sliding variable is chosen as

$$\gamma = \dot{E} + 2lE + l^2\int Edt \qquad (51)$$

where $l \gg 0$.

The $\dot{Y}$ is linearized and is included in equation (50)

$$\dot{E} = V + U\Psi - \dot{Y}_{ref} \tag{52}$$

$$\gamma = V + U\Psi - \dot{Y}_{ref} + 2lE + l^2 \int E dt \tag{53}$$

$$\dot{\gamma} = \dot{V} + \dot{U}\Psi + U\dot{\Psi} - \ddot{Y}_{ref} + 2l\dot{E} + l^2 E \tag{54}$$

Through the equation (8), (54) the control Law for steering angle d can be obtained as

$$d = \frac{1}{\frac{c_{ft}}{M}}\left(-\Lambda|\gamma|^{\frac{p}{q}} sign(\gamma) - U\dot{\Psi} + \ddot{Y}_{ref} - 2l\dot{E} - l^2 E + \frac{c_{rt}+c_{ft}}{MU_0}V + \left(\frac{bc_{rt}-ac_{ft}}{MU_0} - U_0\right)r - \dot{U}\Psi\right) \tag{55}$$

Λ>0 used for the optimal convergence of the sliding variable to zero. The above control law converges the sliding variable in a faster in finite time and with satisfying the reachability condition $\dot{\gamma}\gamma <= -\eta|\gamma|^{\frac{p}{q}}|\gamma|$. When the sliding variable is zero the error also converges to zero and Y starts tracking the reference trajectory at the finite time in real time without having any chattering effect. The model for path is shown below.

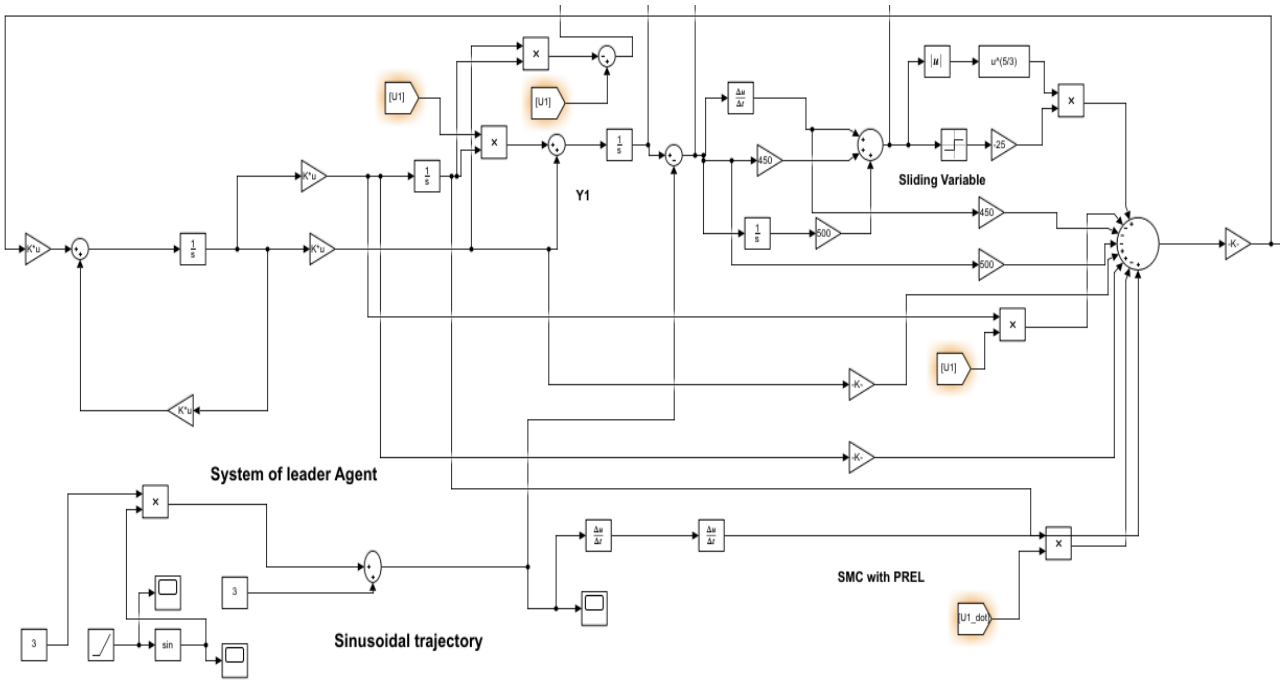


Figure 6 SMC PREL Lateral Control

The power rate exponential reaching law is more robust, chattering free and faster than the conventional reaching law. But there is another disadvantage when there is a faster control

response there might be a possibility of jerk in the vehicle system and the tracking is also fast in real time but Y reaches Y_{ref} asymptotically even though $l \gg 0$ means the error will be very much near to zero and so error is very negligible which can be considered zero practically. So to have a finite time reaching of tracking and to avoid jerk in the vehicle system a more robust sliding mode control should be used for safety purposes and for attaining a platoon which satisfy our need.

III. Super-twisting Sliding Mode Control

It is one of the efficient higher order sliding mode control(HOSMC). For a higher order sliding mode control if a rth order sliding mode is chosen it should be satisfying the condition of lower order dynamics reaching zero. The condition is as follows.

$$s = \dot{s} \dots \dots . = s^{n-1} = 0$$

where $n = 1,2,3$ .... the HOSMC's main aim is to make $n-1$ derivative converge to zero so that a robust sliding mode control with chattering free response is achieved. The chattering effect is early eliminated as all the dynamics of the sliding variable reaches zero due to which the discontinuity effect will become negligible. The decreasing of chattering effect is proportional of the order taken into consideration. If increasing the order of the sliding mode dynamics a decreasing effect of sharp oscillation occurs. For second order HOSMC the task is bring $s, \dot{s}$ to zero in finite time and according to sliding variable chosen the states would reach their desired value in finite time.

Super-Twisting algorithm is can be chosen for relative degree one and two sliding mode which means that the control signal appears in first or second derivative of the sliding variable. The order of the sliding mode is determined by the discontinuity function appearing in the respective derivative of the sliding variable. The super-twisting algorithm is chosen as it is smooth in controlling the sliding variable and the states of the system. In conventional and PERL SMC the sliding variable is brought to zero with the help of the discontinuity in the control signal which can cause a sudden oscillation and jerk to the system. But in super-twisting as a higher order derivative is chosen the control signal is continuous due to which the control action is smooth. The dynamics and the formulation of super-twisting algorithm is represented as

First order derivative of the sliding variable have the dynamics

$$\dot{s} = -\Lambda |s|^{\frac{p}{q}} sign(s) + ß \qquad (56)$$

$$\dot{ß} = -\alpha sign(s) \qquad (57)$$

where $\Lambda, ß > 0$ and chosen according to the response needed. The characteristic of the dynamics above is represented in the below figure.

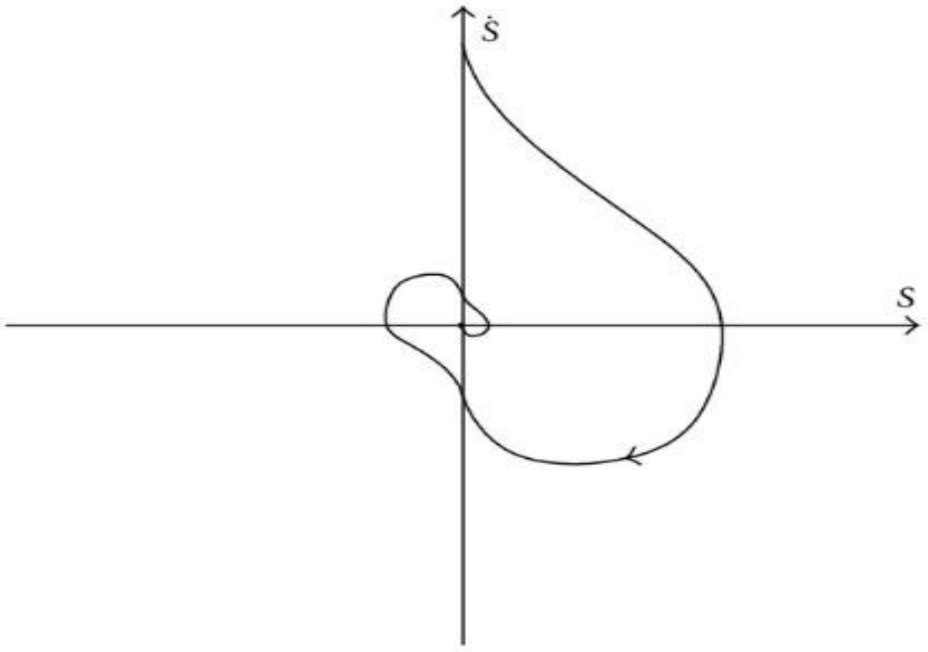


Figure 7 Super-twisting characteristic

As we can see how the sliding variable and its dynamics smoothly reaches zero without any deviation in its path. The second order derivative have the dynamics as

$$\ddot{s} = \frac{-\Lambda \dot{s}|s|^{\frac{p}{q}} - \frac{sp}{q}|s|^{1-\frac{p}{q}}|\dot{s}|}{|s|} - \alpha sign(s) \quad (58)$$

Using the above dynamics the control algorithm is formed for accurate results.

1. Application Super-twisting SMC for Longitudinal Control

A Leader vehicle

The sliding variable is chosen as the error between reference velocity U_{ref} and the present velocity of the leader. The second order sliding mode with relative degree one is chosen for a smooth control. Through the second order dynamics a continuous control signal is represented obtained. The equation is as follows

$$S_1 = U_1 - U_{ref} \quad (59)$$

$$\dot{S_1} = \dot{U}_1 - \dot{U}_{ref} \quad (60)$$

The dynamics of the sliding variable is chose as (56)

$$\dot{S_1} = -\Lambda |S_1|^{\frac{p}{q}} sign(S_1) + ß \quad (61)$$

So the second derivative dynamics can be written as

$$\ddot{S_1} = \frac{-\Lambda \dot{S_1}|S_1|^{\frac{p}{q}} - \frac{S_1 p}{q}|S_1|^{1-\frac{p}{q}}|\dot{S_1}|}{|S_1|} - \alpha sign(S_1) \qquad (62)$$

and the derivative of the sliding variable is taken through equation (7) and

$$\ddot{S_1} = \left(\frac{2U_{01}(FK_l - K_d)\dot{U_1}}{M} + \frac{\dot{f_1}}{M}\right) - \ddot{U}_{ref} \qquad (63)$$

the equation (63) can be expanded as

$$\ddot{S_1} = \left(\frac{2U_{01}(FK_l - K_d)}{M}\left(\frac{2U_{01}(FK_l - K_d)U_1}{M} + \frac{f_1}{M}\right)\right) + \frac{\dot{f_1}}{M} - \ddot{U}_{ref} \qquad (64)$$

through the above equation the dynamics of the control signal is calculated and by integrating it a continuous control signal is activated which is then passed through the system. By equating (64) and (63) the equation of control algorithm can be formed.

$$\dot{f_1} = M\left(\frac{-\Lambda \dot{S_1}|S_1|^{\frac{p}{q}} - \frac{S_1 p}{q}|S_1|^{1-\frac{p}{q}}|\dot{S_1}|}{|S_1|} - \alpha sign(S_1) + \ddot{U}_{ref} - \frac{2U_{01}(FK_l - K_d)}{M}\left(\frac{2U_{01}(FK_l - K_d)U_1}{M} + \frac{f_1}{M}\right)\right)$$

(65)

The control signal obtained is robust and makes the leader agent to track or reach the reference velocity so that the behaviour of the platoon would be satisfactory as the entire platoon characteristic depend on the leader agent. The model of leader agent control is shown below.

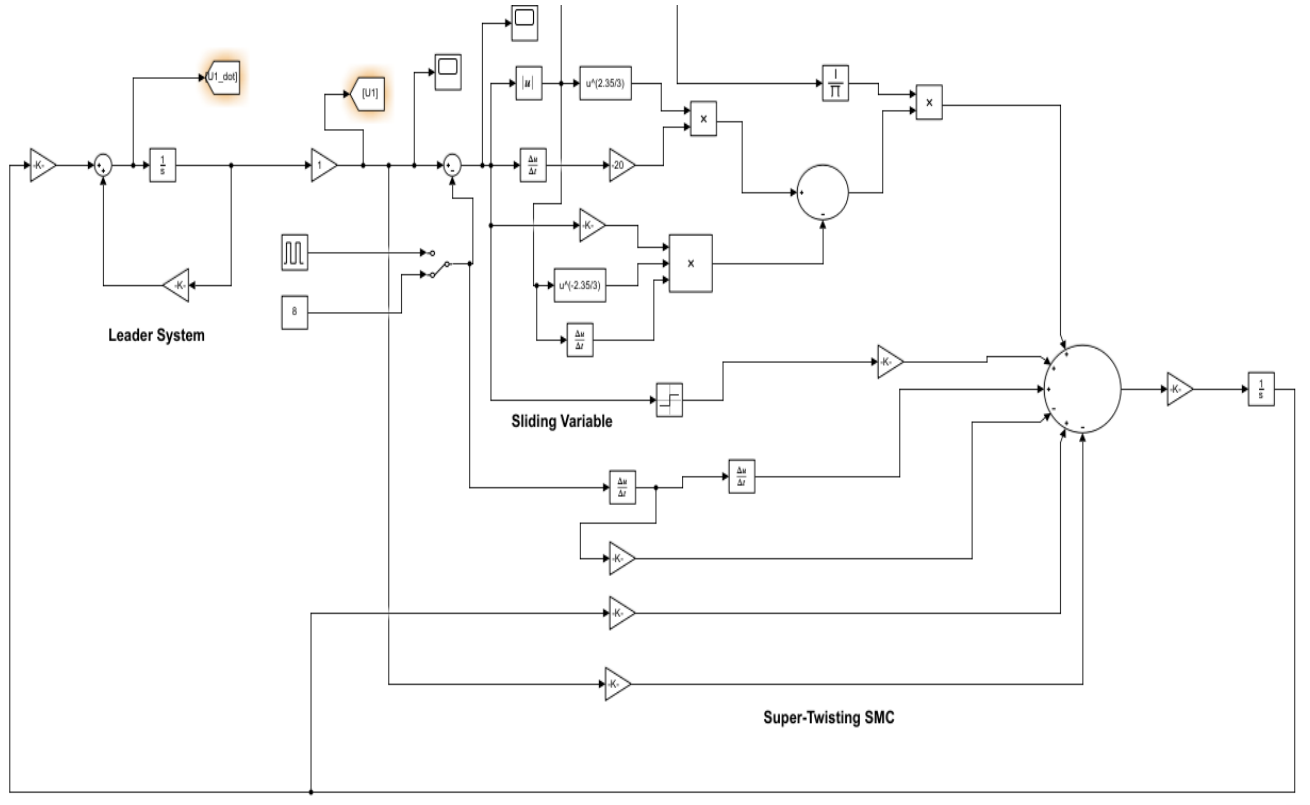


Figure 8 Super-twisting SMC Leader

B)Follower Agent

Here similar to other controller the error is taken between desired safe distance D (41) and the current distance between each agent. Here also a second order with relative degree one super-twisting control method is used. The control signal f_j in which $j = 2 \dots N$ obtained is continuous in nature. The sliding variable is considered as the error. The equations are as follows

$$s_j = D - \left(X_{j-1} - X_j\right) \quad (66)$$

$$s_j = d + t_h U_j - X_{j-1} + X_j \quad (67)$$

$$\dot{s}_j = t_h \dot{U}_j - \dot{X}_{j-1} + \dot{X}_j \quad (68)$$

The dynamics of the first derivative and the second derivative is chosen from (56),(58) respectively. Through (68) the second order derivative can be calculated

$$\ddot{s}_j = t_h \left(\frac{2U_{0j}(FK_l - K_d)}{M}\left(\frac{2U_{0j}(FK_l - K_d)U_j}{M} + \frac{f_j}{M}\right) + \frac{\dot{f}_1}{M}\right) - \ddot{X}_{j-1} + \ddot{X}_j \ (69)$$

By equating (69) with (68) the dynamics of the control signal can be arrived which is then integrated to obtain a continuous control signal which passed through the system.

$$\dot{f}_j = M/t_h\left(\frac{-\Lambda\dot{s}_j|s_j|^{\frac{p}{q}} - \frac{s_j p}{q}|s_j|^{1-\frac{p}{q}}|\dot{s}_j|}{|s_j|} - \alpha sign\left(s_j\right) + \ddot{X}_{j-1} - \ddot{X}_j - \frac{2U_{0j}(FK_l - K_d)}{M}\left(\frac{2U_{0j}(FK_l - K_d)U_j}{M} + \frac{f_j}{M}\right)\right) \quad (70)$$

Through (70) a continuous control algorithm can be used through which a robust control of platoon formation is achieved and with the help of this controller the longitudinal velocity of each agents becomes U_{ref}. Due to f_j a smooth control action is evolved along with negligible chattering effect which does not cause any jerk in the system. The string stability is also maintained due to the careful selection of t_h for the platoon. The model of the follower agent is represented below.

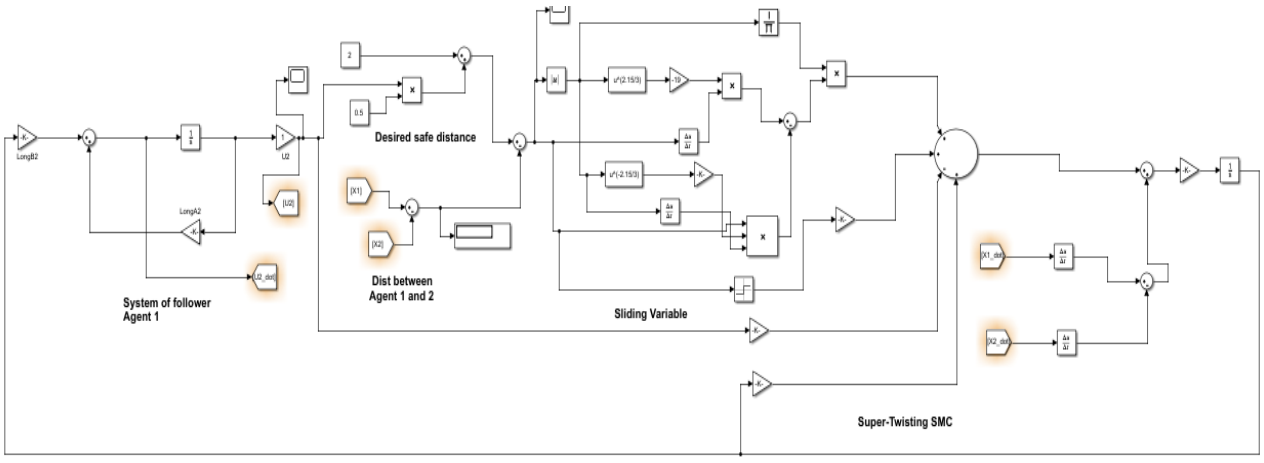


Figure 9 Super-twisting SMC Follower control

2. Application Super-twisting SMC for Lateral Control

For lateral control like the previous sliding mode control the control a sinusoidal reference trajectory Y_{ref} is set and error between the current path of the vehicle Y (j is omitted for simplicity) is calculated. The error is considered as the sliding variable. Here the super-twisting of control is generated through second order sliding mode control and with relative degree 2 so here the control signal appears in the second derivative of sliding variable. The equations are as follows.

$$\gamma = Y - Y_{ref} \qquad (71)$$

$$\dot{\gamma} = \dot{Y} - \dot{Y}_{ref} \qquad (72)$$

$$\ddot{\gamma} = \dot{V} + \dot{U}\Psi + U\dot{\Psi} - \ddot{Y}_{ref} \qquad (73)$$

The second order dynamics of the sliding variable is taken from (58) and it is represented as

$$\ddot{\gamma} = \frac{-\Lambda\dot{\gamma}|\gamma|^{\frac{p}{q}} - \frac{\gamma p}{q}|\gamma|^{1-\frac{p}{q}}|\dot{\gamma}|}{|\gamma|} - \alpha sign(\gamma) \qquad (74)$$

Through equation (8) the steering control from the lateral velocity equation can be computed by substituting it in (73). The (74) is equated with (73) for obtaining the control signal. The control signal obtained is represented below as

$$d = \frac{1}{\frac{c_{ft}}{M}}\left(\frac{-\Lambda\dot{\gamma}|\gamma|^{\frac{p}{q}} - \frac{\gamma p}{q}|\gamma|^{1-\frac{p}{q}}|\dot{\gamma}|}{|\gamma|} - \alpha sign(\gamma) - U\dot{\Psi} + \ddot{Y}_{ref} + \frac{c_{rt}+c_{ft}}{MU_0}V + \left(\frac{bc_{rt}-ac_{ft}}{MU_0} - U_0\right)r - \dot{U}\Psi\right) \qquad (75)$$

The above controller makes the sliding variable reach zero smoothly in finite time without any deviation. As the sliding variable reaches zero in finite so the erro is zero in finite time and Y reachs Y_{ref} in finite time. The figure 10 express the model of later control using Second order super twisting algorithm.

The application of super twisting algorithm in both longitudinal and lateral control would make the sliding surface reach zero without the effect of a discontinuity in finite time. The control signals derived f_j, d does not effect the actuator so there is lower chance of getting into accident due controller. In both longitudinal and lateral control the sliding variable reaches zero in finite time and the error also reaches zero in finite time so a desired platoon formation and path tracking has been achieved which is not susceptible to disturbance and noise. But super twisting has a disadvantage, it is slower than the power rate reaching law. The results of super twisting algorithm would be discussed in the coming chapters.

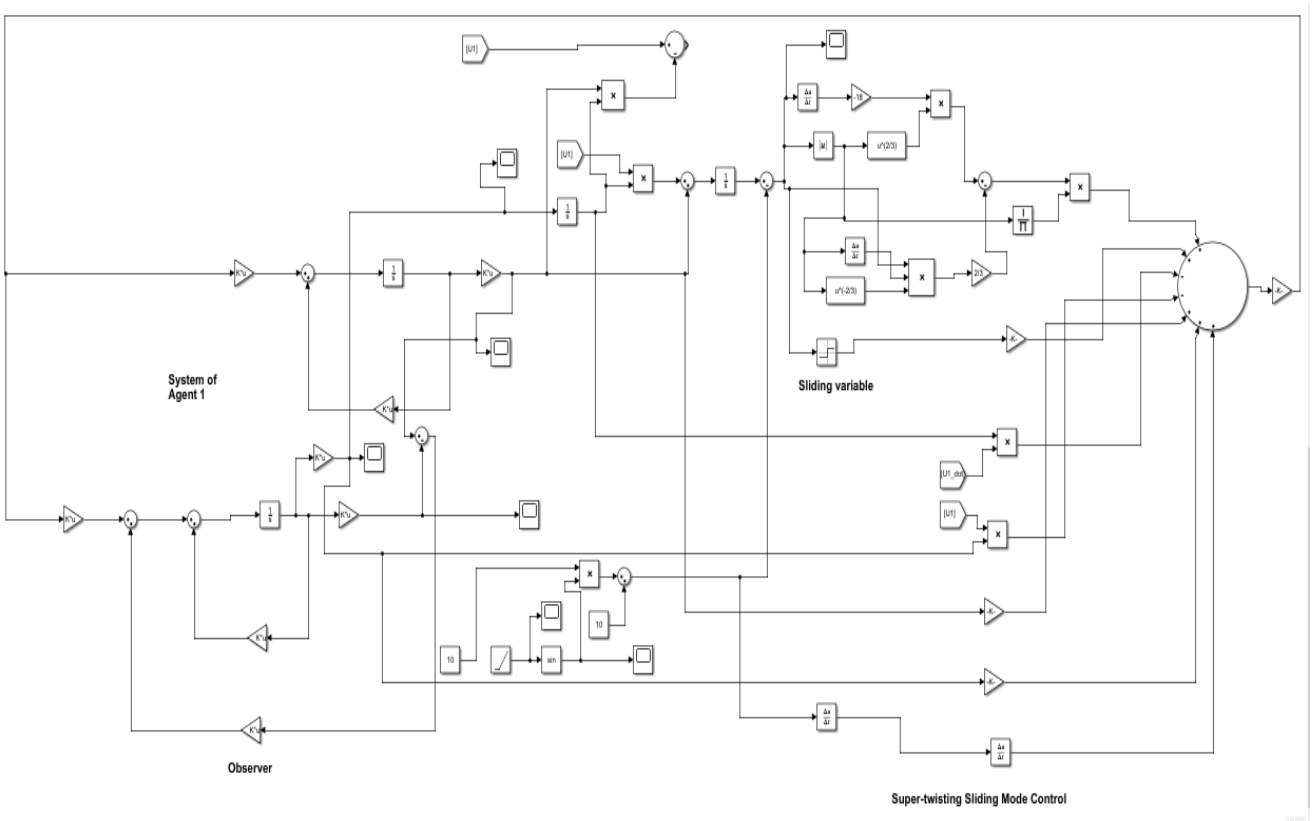


Figure 9 Super-twisting Lateral Control

In practical control system of a self-driving car the yaw rate measurement is not measured directly as the sensors are very expensive,but they are observed through kalman-filter or any observer using the lateral velocity. In this chapter Luenberger observer was used in which the lateral velocity is used as the measured quantity and it is compared with output of the observer calculating the error. The observer using the control signal reduces the error to zero between the two outputs and an accurate yaw rate is acquired from the observer.

Comparative Study Chapter 6

The Control algorithms presented above have some advantages and disadvantages over each other. Like PREL reaching law has faster reaching than Super-twisting where as the later has more stable convergence of sliding variable to zero. To check how each control algorithm have performed with respect to others a performance criteria has to be checked. The criteria contains calculation of Integral absolute error (IAE), Integral time absolute error(ITAE), Integral square error (ISE), Integral time square error(ITSE). The criteria is defined in order to study the error from zero to infinity and by tracking the error minimisation of the performance indices isdone which eventually minimize the error.

IAE criteria:-By considering the absolute error there is an increment in the expression if a error has been detected. The expression for IAE is

$$\text{IAE}=\int_0^{inf} |e(t)|dt$$

ISE criteria:- This criteria is more selective as it squares and integrates each time an error is detected this means that even a small error cannot be ignored. The expression is as follows

$$\text{ISE}=\int_0^{inf} e(t)^2 dt$$

ITAE:- It is more effective than IAE or ISE as it detects longer duration of error as the absolute error is multiplied by time and incremented. The expression is represented below

$$\text{ITAE}=\int_0^{inf} t|e(t)|dt$$

ITSE:- It's working is as similar as ITAE. Here the squred error is multiplied with time and increment each time a error is detected. The formula for ITSE is

$$\text{ITSE}=\int_0^{inf} te(t)^2 dt$$

The table for performance indices is represented as follows

TABLE II: PERFORMANCE INDICES

		Longitudinal				Lateral			
		IAE	ISE	ITAE	ITSE	IAE	ISE	ITAE	ITSE
Leader Agent	Super Twisting	0.504	0.653	0.281	0.007	5.14	37.88	1.73	9.02
	PID	0.44	0.26	0.3	0.03	1.227	4.49	0.32	0.39
	SMC	0.95	0.27	4.17	0.512	4.31	9.294	3.86	6.36
	PREL	0.12	0.19	0.004	0.004	0.78	2.56	0.05	0.12

Agent 2	Super-twisting	6.14	15.49	2.944	3.44	5.115	38.09	0.96	8.01
	PID	1.69	15.24	1.67	4.53	1.351	5.217	0.39	0.45
	SMC	1.536	5.71	7.23	2.08	3.78	7.82	3.09	4.796
	PREL	0.5	1.24	0.023	0.075	0.91	3.24	0.08	0.2
Agent 3	Super-Twisting	7.53	17.69	30.69	18.09	5.08	38.67	0.95	8.54
	PID	1.65	14.83	0.472	6.3821	1.421	5.286	0.537	0.51
	SMC	2.67	4.74	6.34	5.94	3.78	7.7	3.08	4.6
	PREL	0.37	1.5	0.011	0.03	1.002	3.56	0.08	0.27

The table above represented the values for each criteria for different controller for 3 different agents. The rule of the criteria is that if there are minimum values in all the criteria then the performance of the controller is satisfactory. As we can see from the table PREL reaching law has minimum values compared to others which shows the PREL has performed more satisfactorily than others. But we even have to compare the control energy through which we can know the efficient controller.

TABLE III:CONTROL ENERGY

		Longitudinal Control	Lateral Control
Agent 1	Super-twisting	$4 * 10^7$	51.2
	PREL	$3.13 * 10^8$	254
	PID	$8.84 * 10^8$	814
	SMC	$6.89 * 10^9$	0.914
Agent 2	Super-twisting	$1.2 * 10^8$	54.09
	PREL	$8.38 * 10^9$	197
	PID	$1.213 * 10^{10}$	384
	SMC	$7.815 * 10^{10}$	1.078
Agent 3	Super-twisting	$1.77 * 10\text{^}8$	52.3
	PREL	$3.19 * 10^{10}$	195.2
	PID	$5.5 * 10^{10}$	392
	SMC	$4.9 * 10^{10}$	1.078

The above table represents the control energy which specify how much energy is consumed by the control output to bring the states from current value to the desired value. As we can see

form the table that super-twisting control has consumed less energy compared to others. As super-twisting second order control is continuous control and more accurate than PID.

In this chapter a comparison of performance and energy consumption has been discussed but the results of the controller has to compared to reach a conclusion of which controller has performed best. The following chapter analyses results of each controller through which a better controller can be chosen for urban platooning.

Results and Discussions Chapter 7

The automation in platooning is conducted by applying a robust longitudinal and lateral control of the agents or vehicles. The platoon contains a leader vehicle and two follower agents. The platoon should be tracking a reference trajectory of Y_{ref} which is a lateral position of the platoon and should also reach a reference longitudinal velocity considered as 8m/s.In this chapter results of each and every controller is presented. Each controller's performance is discussed through their results and they are compared with each other for concluding the most robust and accurate controller. The initial conditions are of the position, velocity and yaw angle are considered as $x_0 = 50m, x_1 = 35m, x_2 = 20m, Y_0, Y_1, Y_2$ are $0m, U_l = 10m/s, U_1 = 5m/s, U_2 = 1m/s, V_0, V_1, V_2$,yaw for all the agents are $0m/s$ and $0rad$ respectively, $l = 10$. For the observer the desired poles are taken as -10,-10. The power rate reaching law and Supertwisting algorithm contains the parameters Λ and $\frac{p}{q}$ where $\Lambda > 0$ and chosen as 25. The condition $0 < \frac{p}{q} < 1$ should be satisfied so that the convergence of the sliding variable be faster and for removing the effects of chattering. It is chosen as $p = 2, q = 3$. The gain Λ and $|s|^{\frac{p}{q}}$ ensure faster reaching and slower convergence when states are much more further and nearer to its desired value respectively. If $\frac{p}{q}$ id near 1 then sliding variable s does not converge to zero and the chattering effect is increased and vice versa when $\frac{p}{q}$ is near zero. The reaching law and super-twisting with the above mentioned conditions secure the robustness and elimination of the chattering which makes sure that system states remain in their desired value without deviation. For PID longitudinal controller the $K_p = 5000,7500,8500$, $K_i = 5500,8000,9000$ and $K_d = 15,30,40$ for Agent 1,2,3 respectively and for PID lateral$K_p = 22,22,22$, $K_i = 100,100,100$ and $K_d = 2,2,2$ for Agent 1,2,3 respectively. The results of the simulation are shown in following figures.

The results below brings a contrast between SMC,PREL SMC, Super-twisting SMC and PID. Fig 10,11,12,13 shows the lateral control for each controller, as it is shown that in PID the controller is much slower than other controllers. There is an overshoot in PID which can cause sudden jerk in the system and this over shoot is also present in PREL but its time is lesser than PID. Conventional SMC used is faster than PID but it is having a chattering effect which can cause damage to the actuators of the system. It is clear that SMC is working better than PID as in PID there are some large occilations near zero which is not present in SMC. The PREL has

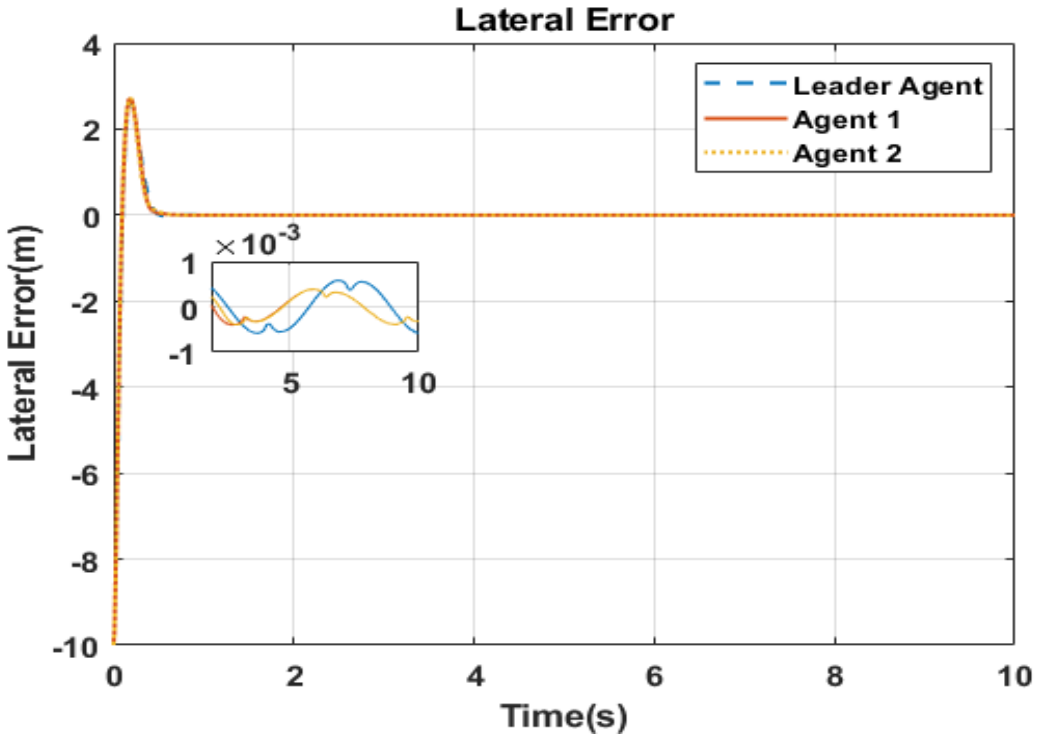


Figure-10 Lateral Error PREL

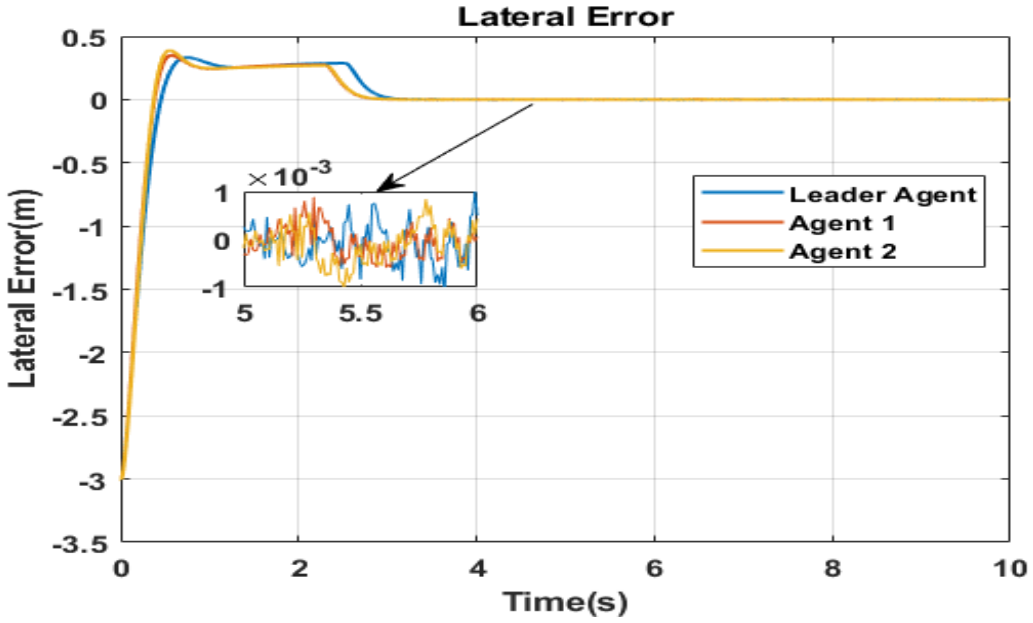


Figure 11 lateral Error SMC

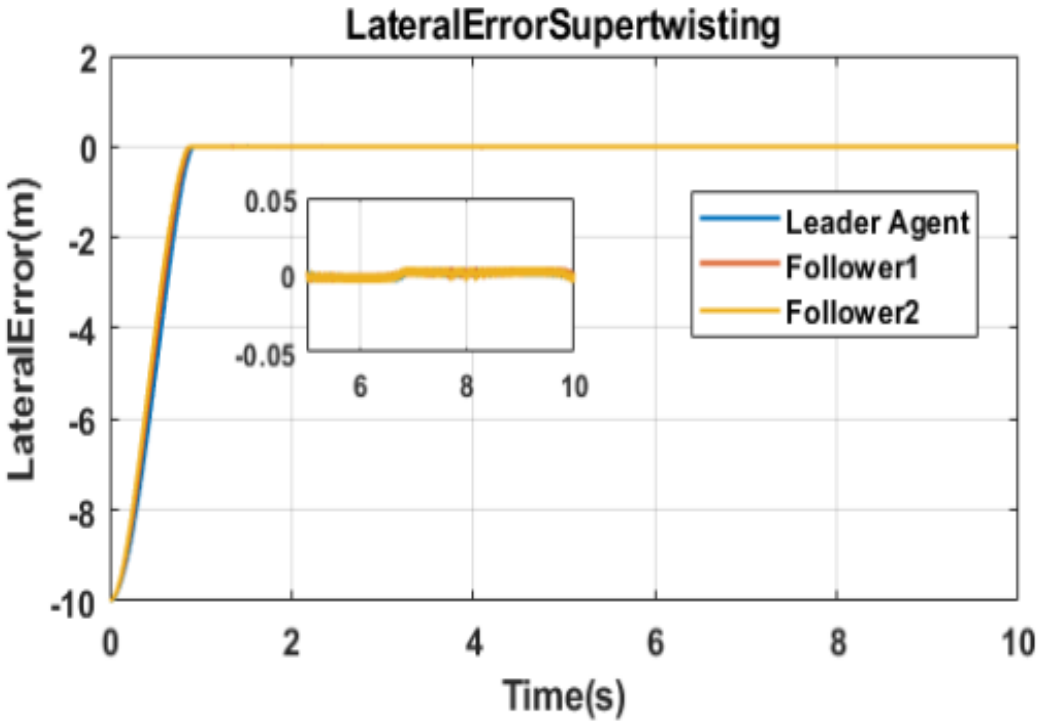


Figure-12 Lateral Error Super-twisting

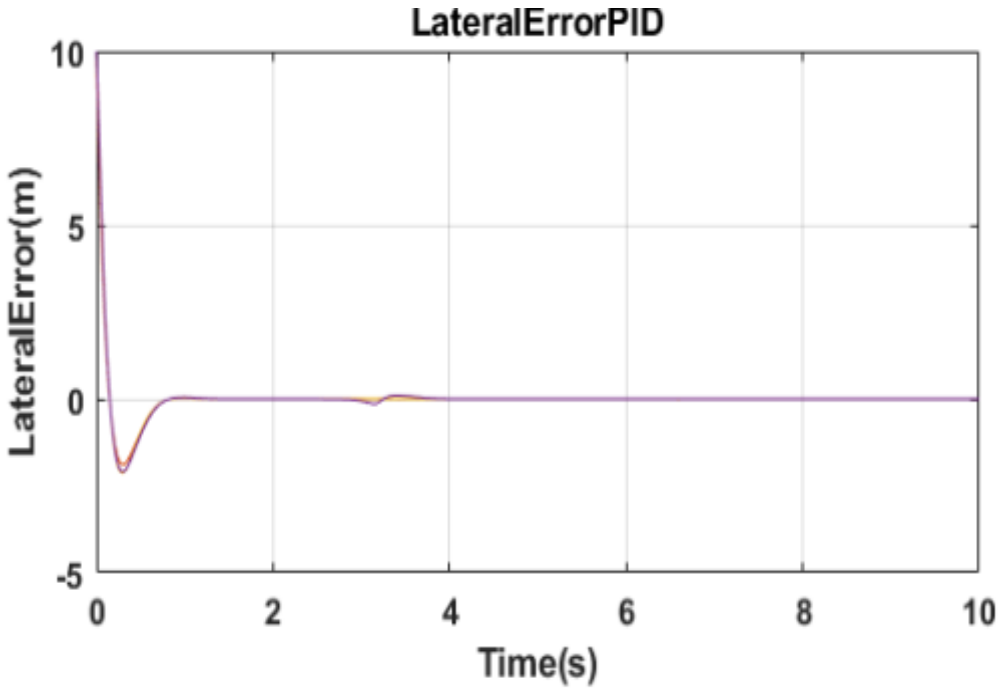


Figure-13 Lateral Error PID

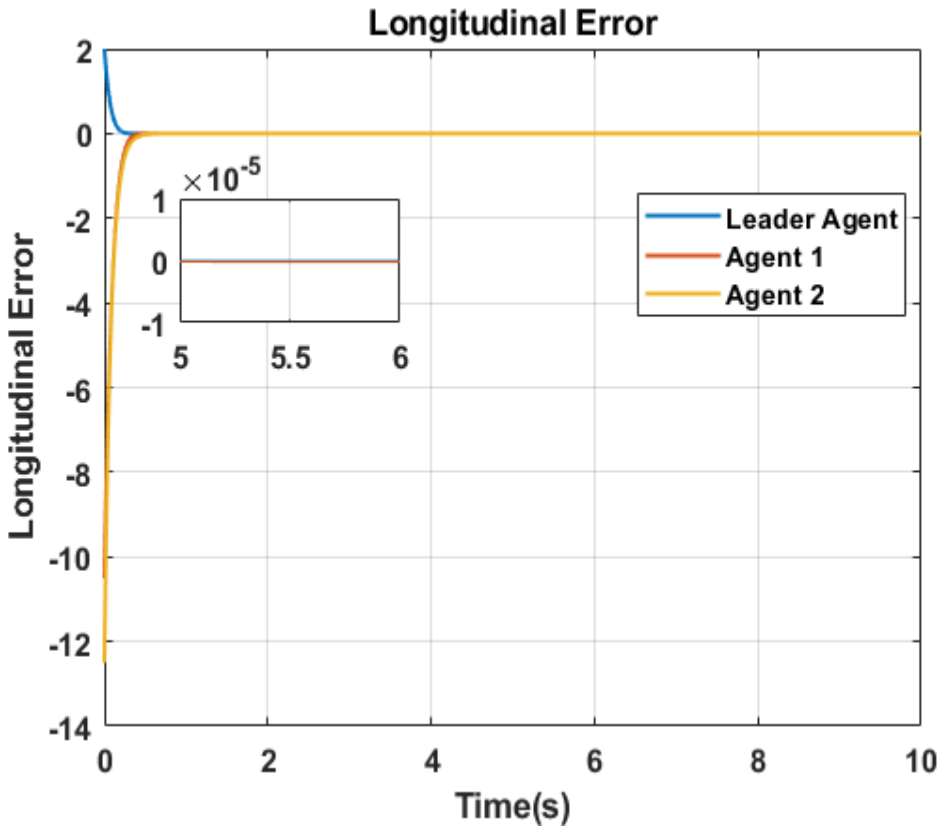


Figure-14 Longitudinal Error PREL

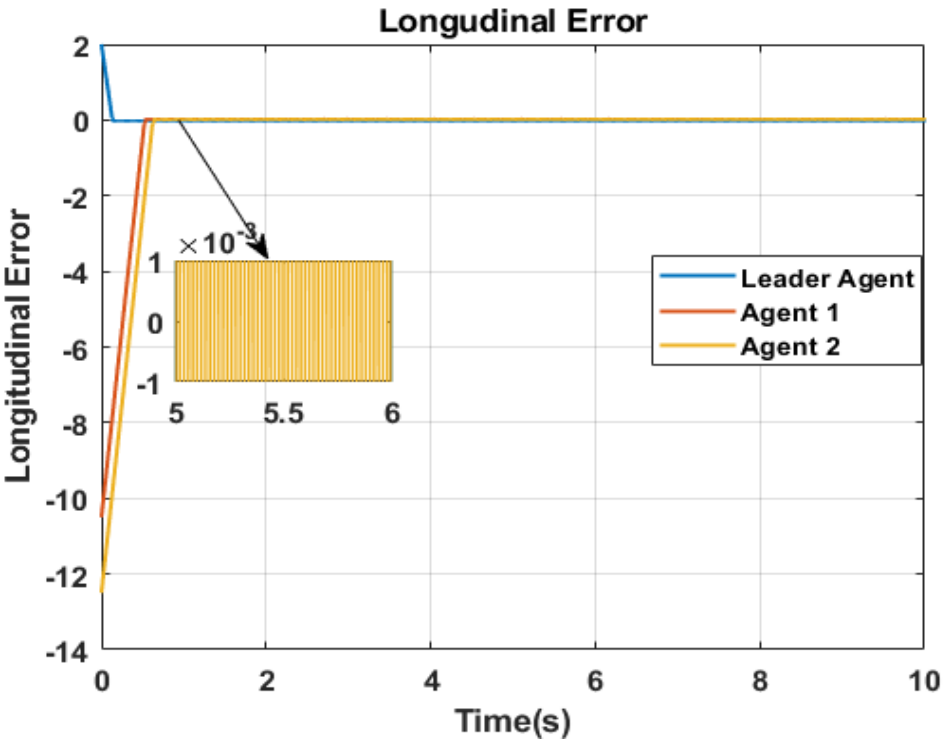


Figure-15 Longitudinal Error SMC

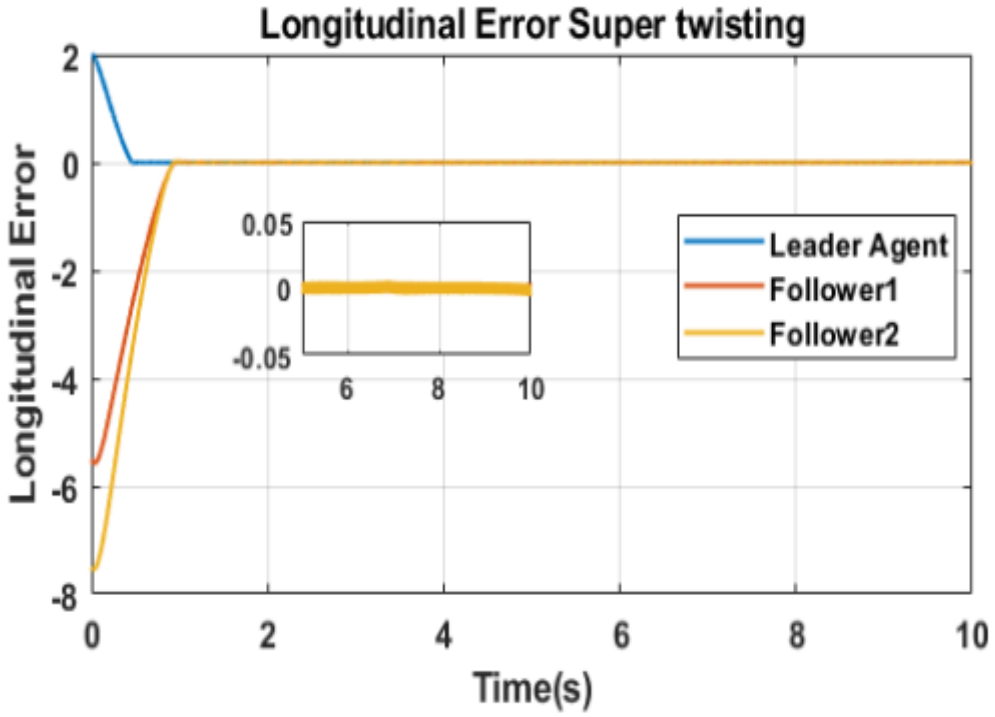


Figure-16 Longitudinal Error Supertwisting

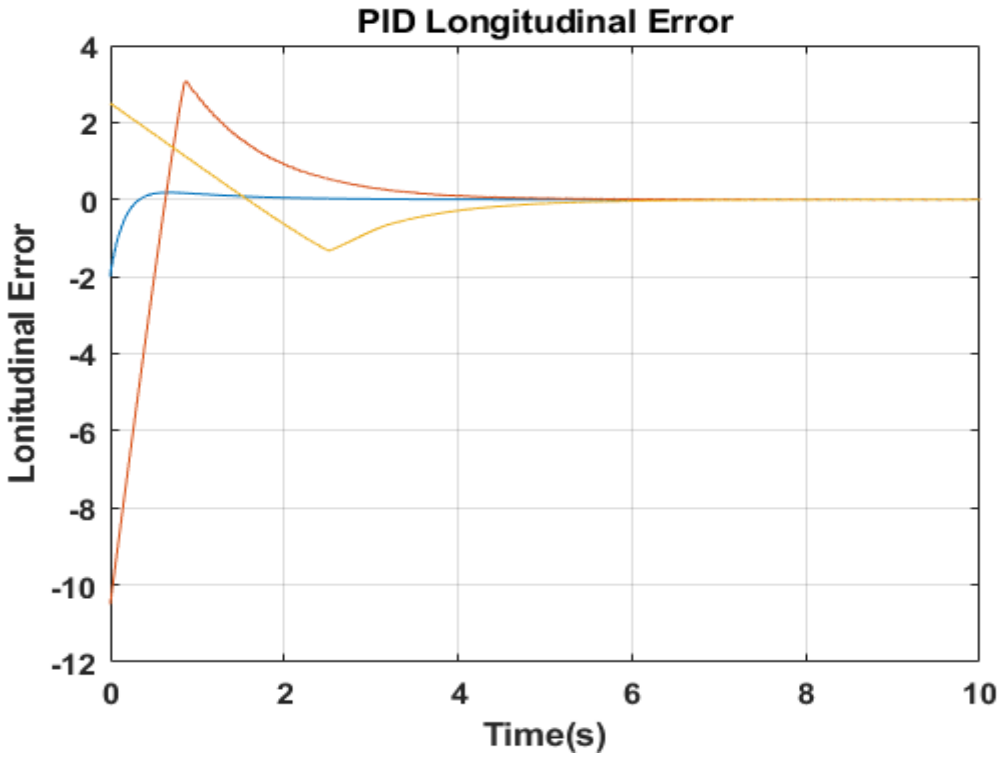


Figure-17 Longitudinal Error PID

performed better than PID and SMC. It neither have chattering effect not large occilations near zero. It is much faster than other controllers but it has an overshoot due to the PID sliding surface which causes jerk in the actuators. The chattering effect is completely eliminated due to its dynamics of the sliding variable and proper selection of the reaching parameters. Due to these careful selection of the parameters the controller energy is lesser than both SMC and PID controller. It is also better than SMC as it is having larger overshoot than PREL due to the selection of the sliding surface. The super-twisting controller has performed much better than SMC, PID and PREL. First in supertwisting Y reaches Y_{ref} in finite time without having any overshoot or jerk. Due to this it can be considered more safer than PREL and other controllers. But there are chattering effects but it is lesser than conventional SMC due to its continuous control signal. The results also shows that there is not even a slightest oscillation near zero. Even though it is slightly slower than PREL its energy consumption is very less and the controller output is smooth.

Figure 14,15,16,17 shows the performance of each controller for longitudinal control. As it is shown that PID has under-performed even though a large gain in its system so from its performance it is not suitable for the longitudinal control. The SMC has performed much better than PID but its chattering effect is large and can easily dame the actuators of the system even though it is faster in reaching zero. PREL control is much faster than SMC, Supertwisting and there are no chattering effects. Super-twisting algorithm is faster than SMC but slower than PREL and its chattering effect is much lesser than SMC. Here also the energy consumption is lesser than other controller and it is lot smoother in nature.

Conclusion and Future Scope Chapter 7

I. Conclusion

The reported in this thesis is the application of PID, conventional sliding mode control, power rate exponential reaching law and Super-twisting second order sliding mode controller has been discussed which is then used to propose a robust sliding mode control for platoon control and formation. First the mathematical model of the vehicle system has been discussed which is later used as agents for platooning. The mathematics of each controller and its application in the model has been discussed extensively and the reason of application of each controller is represented.

The results of each controller has been discussed in detail to understand the suitability of these controllers. The proposed control schemes which are power rate reaching law and Super-twisting sliding mode control satisfies the string stability condition of the platoon with finite time faster convergence of states to its reference point, trajectory and it is use-full in real time application of sensor fusion in self driving cars. The super-twisting control can be recommended as best for platooning chattering effects a little higher than PREL as it has proven to be smooth and more energy efficient in nature though it has chat The control schemes also prevents the chattering effects which can cause multiple damage and safety issues.

II. Future Scope

The results expressed that power rate reaching law and super-twisting control algorithm are much more better and suitable for application in platooning. But both is having an advantage and disadvantage over each which is chattering effect and a small amount of jerk. There may be a solution for the above problem which is third order super twisting algorithm which can be more efficient than the second order which have a little chattering effect. There might also be a scope of more efficient method for urban platooning like providing a better and efficient communication network between each agent which can provide more data and parameters to be controlled.

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